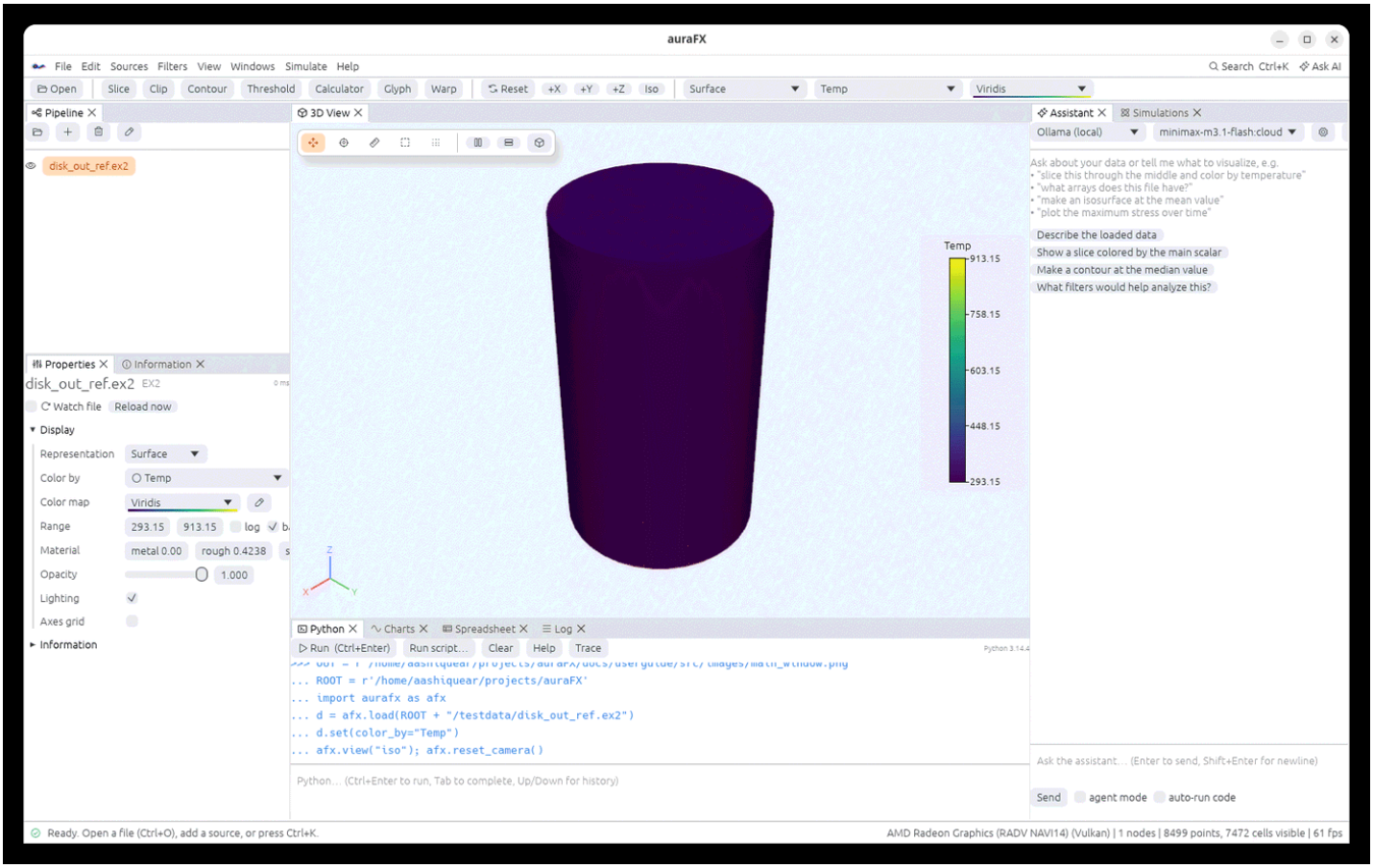
# auraFX User Guide

auraFX is a scientific visualization, analysis and simulation workbench. It opens the files your solvers write (Exodus, VTK, OpenFOAM, LAMMPS, CGNS, EnSight, NetCDF and many more), shows them in a 3D view drawn by your GPU, lets you cut, contour, threshold, trace and measure them with the usual filters, and adds three things next to the 3D view:

- a **Python console** with a one-line-per-idea API (`afx.load("run.e").slice().color_by("T")`),
- an **AI assistant** that sees your pipeline and writes runnable code for you,
- a **simulation runner** that launches solvers locally, over SSH or through Slurm and opens the results when they finish.

This guide walks through every feature. Each chapter says where a feature is in the window, how to call it from Python, and what its limits are.



## How this guide is organised

| Part | What it covers |
|---|---|
| Basics | opening files, the pipeline, filters, coloring, views, selection, time and animation |
| Rendering | lighting, shadows, ambient occlusion, ray tracing, materials, volumes, point clouds |
| Scripting | the Python API, the AI assistant, writing plugins |
| Data | every file format read and written, exporting pictures and scenes, running simulations, working on remote machines |
| Reference | settings and keyboard shortcuts |

## Formats of this guide

The guide is kept as Markdown in the repository ( `docs/userguide/src` ), published as an online book, and printed to a PDF ( `docs/userguide/auraFX-user-guide.pdf` ). All three come from the same source files, so they always say the same thing.

# Getting started

## Installing

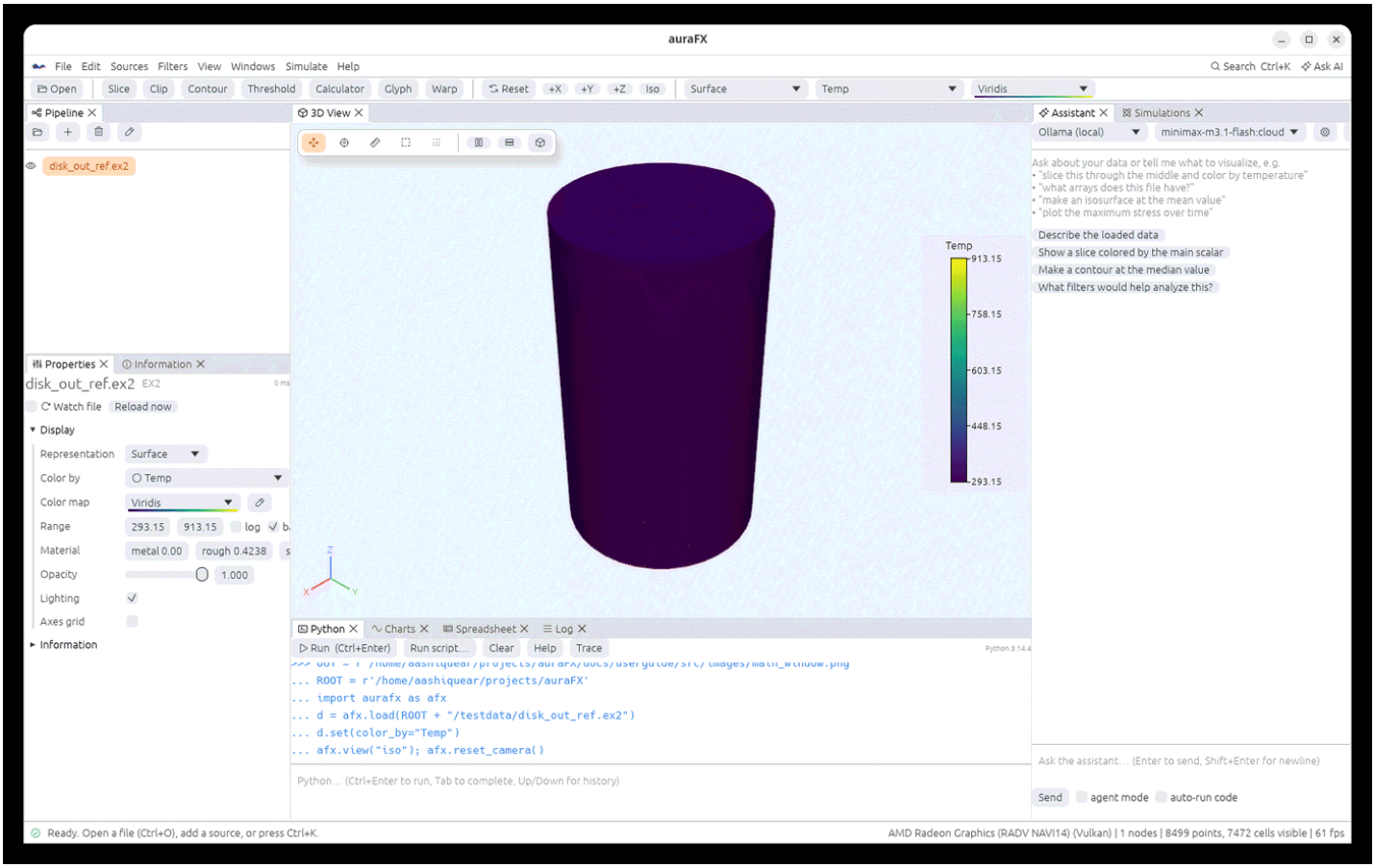
Prebuilt packages for Linux, macOS and Windows are attached to each release, and a one-line installer puts the app in your path:

```
curl -fsSL
https://raw.githubusercontent.com/aashiquear/auraFX/main/scripts/install.sh | sh
# Linux, macOS
irm https://raw.githubusercontent.com/aashiquear/auraFX/main/scripts/install.ps1 |
iex          # Windows PowerShell
```

Desktop packages bundle Python 3.12 with NumPy, so the Python console works without setup. To build from source you need Rust 1.85 or newer, Python 3.10+ with development headers, and a Vulkan (Linux, Windows) or Metal (macOS) GPU driver:

```
cargo build --release
./target/release/aurafx                    # start
./target/release/aurafx run.e              # open a file
./target/release/aurafx --script setup.py # run a Python script at startup
```

# The window




The window has five areas; every panel can be closed, reopened from the **Windows** menu, and dragged to another place.

| Area | Purpose |
|---|---|
| **Pipeline** (top left) | the files you opened and the filters below them; click to select, click the eye to show or hide |
| **Properties** / **Information** (bottom left) | the selected object's parameters and display settings; the Information tab shows arrays, ranges, bounds, cell types and time steps |
| **3D View** (middle) | the picture; drag to rotate, right-drag or wheel to zoom, middle-drag (or Shift-drag) to pan, double-click to fit |
| **Python**, **Charts**, **Spreadsheet**, **Log** (bottom) | the console, plots, the data as a table, messages |
| **Assistant**, **Simulations** (right) | the AI chat and the job runner |

The toolbar has the most used filters (Slice, Clip, Contour, Threshold, Calculator, Glyph, Warp), the camera buttons (Reset, +X, +Y, +Z, Iso) and, for the selected object, its representation, the

array it is colored by and the color map. The status bar at the bottom names the GPU in use and counts the visible points and cells.

## Opening a file

- **File › Open…** (Ctrl+O), or drop a file onto the window, or start the app with the file name.
- **File › Open Recent** keeps the last files; the three sample datasets at the bottom of that menu (a heated disk, a wavelet volume, a sphere) are good for trying things out.
- Numbered files (`run_0001.vtu`, `run_0002.vtu`, …), `.pvd` collections, Exodus and OpenFOAM cases open as time series: the time toolbar appears under the main toolbar.

When a file opens, the object is colored by its first array and the camera fits the data. The file is watched: if a solver rewrites it, auraFX reloads it (**Watch file** in Properties).

```
d = afx.load("run.e")              # Python does the same
d.set(color_by="Temp")
afx.view("iso"); afx.reset_camera()
```

## Applying a filter

Select an object, then pick a filter from the toolbar, the **Filters** menu, or the command palette (**Ctrl+K**, type part of its name). The filter runs at once with sensible defaults (a slice through the center, a contour at the middle value) and appears below its input in the Pipeline. Change its parameters in Properties; they apply as you type. **Re-run** forces a recomputation and **Reset to defaults** restores the defaults.

Filters can be stacked: a Contour of a Clip of a file. Delete an object with the trash button or the Delete key; the objects below it go too.

## Saving your work

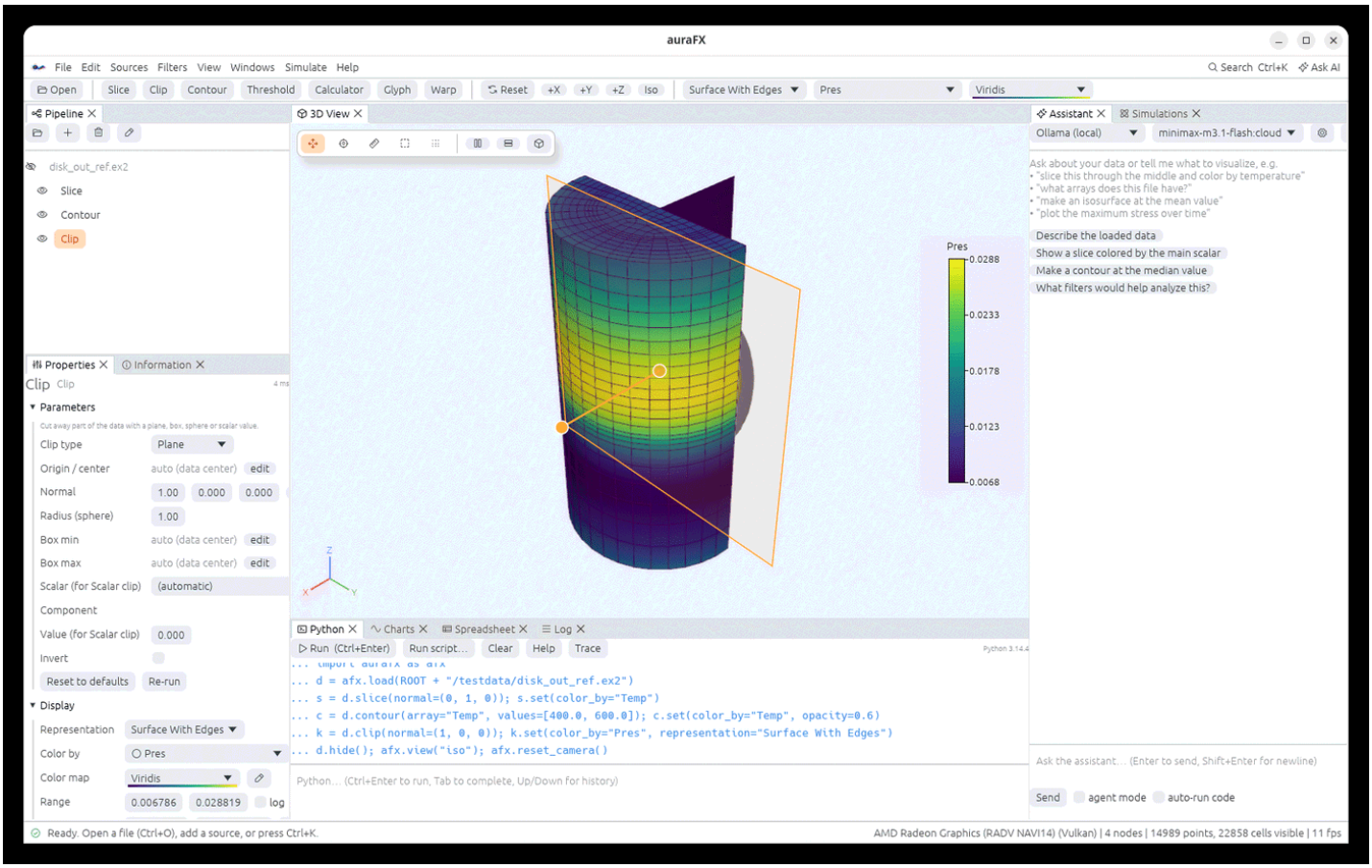
- **File › Save State** (Ctrl+S) writes an `.afx` file holding the pipeline, every parameter, the display settings, the camera and any keyframes. **File › Open** reopens it, including objects that live on a remote server.
- On the next start, **Restore last session** in the empty view brings back the last state.
- Right-click an object › **Save data…** writes its output to a file (see File formats).
- **Edit › Undo / Redo** cover every change to the pipeline and display settings.

# Getting help

- **Help › Keyboard shortcuts** and **Help › Python API cheat sheet** open inside the app.
- The command palette (Ctrl+K) finds filters, sources and actions by name.
- The AI assistant answers questions about the loaded data and the app (see AI assistant).

# Filters

A filter takes the output of the object above it and makes a new object. Select an input, choose the filter, adjust it in Properties. Every filter is also a method in Python ( `node.slice(...)` ), and the assistant knows all of them.




## Common filters (toolbar)

| Filter | What it does | Main parameters |
|---|---|---|
| **Slice** | the data on a plane or a sphere, with the values interpolated onto it | type, origin, normal (or radius); the plane can be dragged in the view |
| **Clip** | keeps one side of a plane, the inside/outside of a box or sphere, or the cells above/below a scalar value; **crinkle** keeps whole cells | type, origin, normal, box corners, radius, scalar and value, invert |

| Filter | What it does | Main parameters |
|---|---|---|
| **Contour** | isosurfaces (or isolines on 2-D data) at one or more values | array, values; computes normals |
| **Threshold** | the cells whose values lie in a range | array, min, max, all/any point |
| **Calculator** | a new array from an expression over other arrays ( `sqrt(u*u+v*v)` , `mag(V)` , vectors with `iHat` …) | expression, result name, point or cell |
| **Glyph** | arrows, cones, spheres, cubes or lines at the points, scaled and oriented by arrays | shape, orientation and scale arrays, factor, maximum glyphs |
| **Warp By Vector / Scalar** | moves points by a vector array (displacements) or along the normal by a scalar | array, factor |
| **Stream Tracer** | streamlines of a vector field from a line or a point cloud of seeds; **Tube** makes them solid | vectors, seed type, seed line or cloud, integration direction and length |
| **Plot Over Line** | samples the data along a line and plots it in Charts | the two end points (draggable) |
| **Probe Location** | the values at one point, shown in Information and the Spreadsheet | location (draggable) |

# Analysis

Resample To Image, Resample With Dataset, Histogram, Descriptive Statistics, Integrate Variables (areas, volumes and integrals of every array), Gradient (with divergence, vorticity and Q-criterion), Cell Size, Mesh Quality, Elevation, Cell Centers, Connectivity, Find Data, Extract Selection, Temporal Statistics (mean, min, max and standard deviation over all time steps), Temporal Interpolator, Temporal Shift Scale, Temporal Array Operator, and the particle filters (Particle Tracer, Particle Path, Streak Line) that follow particles through a time-varying flow. Texture Map to Plane / Cylinder / Sphere, Perlin Noise and Spatio Temporal Harmonics are described below.

# Geometry

Extract Surface, Extract Edges, Feature Edges, Outline, Transform, Reflect, Shrink, Clean, Surface Normals, Smooth, Triangulate, Tetrahedralize, Tessellate (curved cells into straight pieces), Mask Points, Merge Blocks, Extract Block, Extract Subset (a sub-volume of an image), Append Datasets / Geometry / Attributes, Cell Data to Point Data and back, Extract Component, Rename Array, Pass Arrays, and OpenFOAM Boundary Patches.

# Sources

**Sources** adds objects without a file: Sphere, Cone, Cylinder, Box, Plane, Line, Point Cloud, Wavelet (a test volume), Spatio Temporal Harmonics (a test volume of waves with its own time steps), AMR Gaussian Pulse (a small AMR dataset) and Unstructured Cell Types (one of every cell type, for tests).

# Textures and made-up data

## Showing an image on a surface

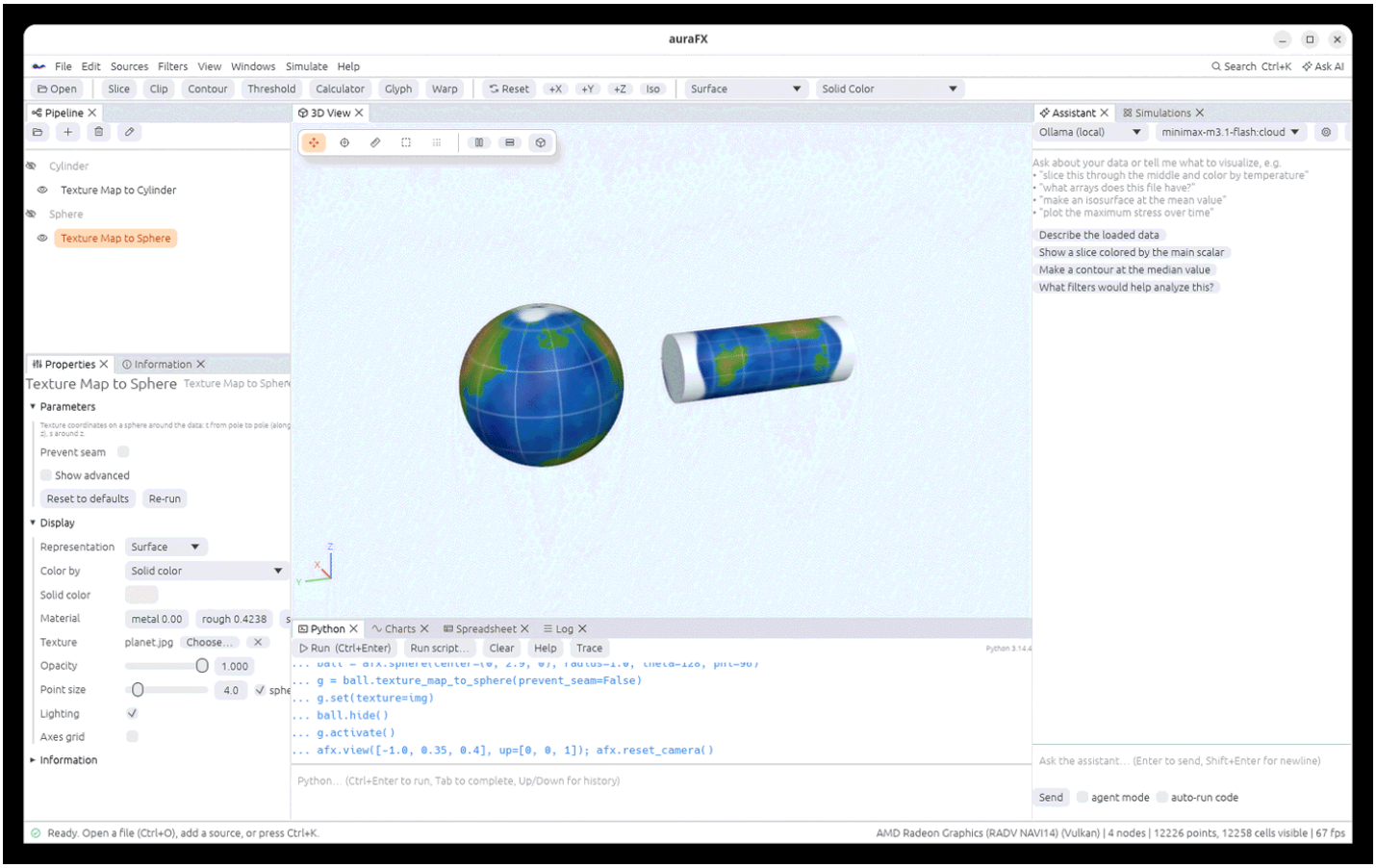
An image (PNG or JPEG) can be wrapped onto a surface, the way a world map goes onto a globe. The surface needs **texture coordinates**, which say where each point falls in the image. Some files and the Plane source come with them. For anything else, add one of these filters (**Filters › Texture Map to Plane / Cylinder / Sphere**):

| Filter | How the image is laid on | Parameters |
|---|---|---|
| **Texture Map to Plane** | projected straight onto the surface from a plane, like a slide projector | **Automatic** fits the plane to the points and stretches the image over the bounds. Turn it off and give **Origin** (image corner), **Point 1** (the image's right edge) and **Point 2** (its top edge). |
| **Texture Map to Cylinder** | rolled around an axis: up the axis is up the image, around it is across | **Automatic** puts the axis along the object's longest side. Turn it off and give **Point 1** and **Point 2**, the axis ends. |
| **Texture Map to** | wrapped around a center: the image's top | **Center** (under Show advanced; empty = the middle of the points) |

| Filter | How the image is laid on | Parameters |
|---|---|---|
| **Sphere** | at the +z pole, its bottom at the −z pole | |

**Prevent seam** (cylinder and sphere, on by default) runs the image across half the way round and back mirrored. That avoids a visible join, but shows the image twice. Turn it off to wrap the image once, as for a world map. A thin seam then shows where the edges meet (ParaView shows the same seam).

Then pick the image in **Properties › Display › Texture › Choose…** (× removes it). The row appears when the object has texture coordinates. The image replaces the array colors and the solid color; lighting and the material still apply. Coordinates outside 0 to 1 repeat the image.




```
ball = afx.sphere(radius=1, theta=128, phi=96)
globe = ball.texture_map_to_sphere(prevent_seam=False)
globe.set(texture="/path/to/world.jpg")        # texture=None removes it
p = d.texture_map_to_plane(automatic=False, origin=(0, 0, 0), point1=(10, 0, 0),
point2=(0, 5, 0))
c = d.texture_map_to_cylinder(point1=(0, 0, -1), point2=(0, 0, 1),
automatic=False)
```

The new array is called **Texture Coordinates**. The three filters give the same numbers as ParaView's filters of the same names, and the coordinates can also be colored by like any other array.

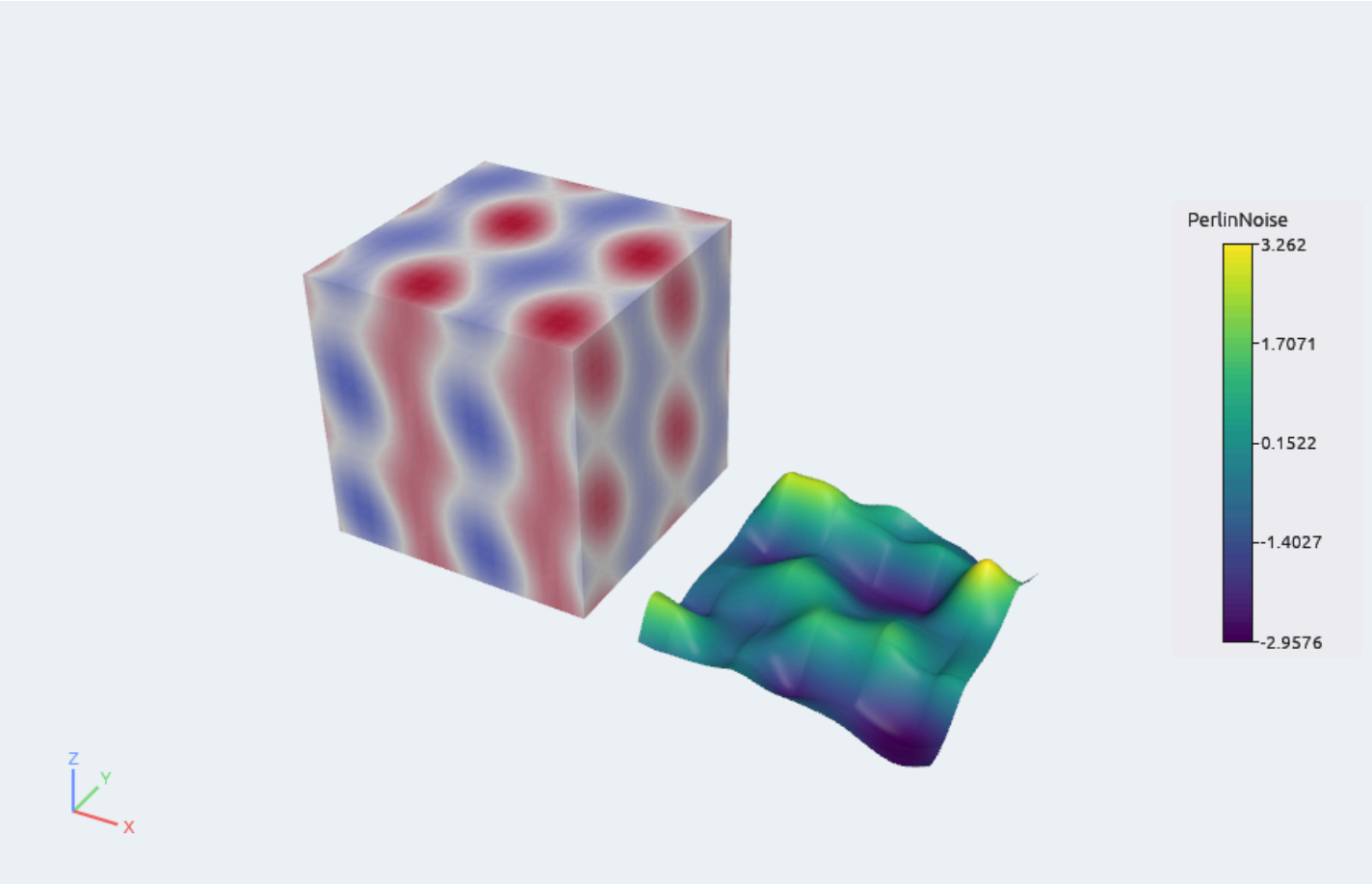
Limits:

- The image is shown on surfaces (Surface and Surface With Edges) and on lines. Points and volumes don't show it.
- Ray tracing, scene export and pictures drawn on a server show the solid color instead of the image.
- The coordinates are stored at half precision for drawing (about 1/2000 of the image width), which only matters for images repeated many times.

## Made-up data for tests and demos

| Filter or source | What it makes | Parameters |
|---|---|---|
| **Perlin Noise** (filter) | a smooth random-looking value at every point (array **PerlinNoise**), for test data or rough-looking surfaces | **Frequency** (bumps per unit along x, y, z), **Phase** (shifts the pattern), **Amplitude**, result name |
| **Spatio Temporal Harmonics** (filter) | a sum of travelling sine waves at every point (array **SpatioTemporalHarmonics**) | **Harmonics**: 6 numbers per wave (amplitude, temporal frequency, wave vector x y z, phase). The value is the sum of amplitude × sin(frequency × time + wave vector · point + phase). |
| **Spatio Temporal Harmonics** (source, **Sources** menu) | an image volume with those waves and **20 time steps of its own** (one period), so it animates without a file | **Whole extent** (x min, x max, y min, y max, z min, z max; spacing 1), **Harmonics**, **Time steps** |

The source's time steps join the time toolbar like a file's. The filter follows the time of its input. On data without time steps it uses time 0, as ParaView does. Both match ParaView's filter and source with the same defaults (ParaView rounds π/5 to 0.6283 in its defaults, and so does auraFX).




```
h = afx.spatio_temporal_harmonics_grid()     # waves in a 21×21×21 volume, 20 time steps
h.set(harmonics=[2, 1, 0.3, 0, 0, 0])         # one wave along x
afx.time(5)
n = afx.plane(xres=200, yres=200).perlin_noise(frequency=(4, 4, 4), amplitude=0.1)
n.warp_by_scalar(array="PerlinNoise")
```

## Programmable filters

**Programmable Filter**, **Programmable Source** and **Python Calculator** run Python that is stored in the pipeline and in state files. The filter's code gets `data` (points and arrays as NumPy) and returns it changed, or a new dataset built with `afx.make_*`. For code you want to keep across sessions, write a plugin.

## Filters with several inputs

Append Datasets, Append Geometry, Append Attributes and Resample With Dataset take several inputs. Ctrl+click the inputs in the Pipeline, then choose the filter; the Properties panel

gets an **Inputs** section where the order can be changed.

```
afx.apply_multi("append_datasets", [a, b, c])
r = probe_mesh.resample_with_dataset(source)
```

## Parameters that say “auto”

Many parameters start as **auto (data center)** or similar: the value is taken from the data each time it runs (the center of the bounds, the middle of a range, 5 % of the size). Press **edit** to type a value instead. Auto values follow the data through time steps and across pieces of a parallel server.

## Speed

Slice, Contour, Clip, Threshold, Extract Surface, Stream Tracer, Gradient and Integrate Variables run on every core and give exactly the same result on any number of threads. A slice of 4 million cells takes about 80 ms. The thread count can be limited in **Settings › Filter threads** (0 = all cores) or with `AURAFX_THREADS`.

## From Python

```
afx.filters()                 # every filter with its parameters
afx.help_filter("clip")       # one filter explained
s = d.slice(normal=(0, 1, 0))
c = d.contour(array="Temp", values=[400, 600])
k = d.clip(type="Box", box_min=(-3, -3, -2), box_max=(3, 3, 4))
t = d.threshold(array="Pres", min=0.01)
s.set(origin=(0, 1, 0))       # change a parameter; the filter re-runs
s.update()                    # wait for it
```

# Coloring and color maps

## Color by an array

The **Color by** box in Properties (and in the toolbar) lists every point and cell array of the object. Vectors are colored by magnitude unless a **Component** is chosen. **Solid color** uses one color for the whole object. New files are colored by their first array automatically.

**Range** is taken from the data; type values to fix it (useful when comparing time steps or objects), press **auto** to return to the data range, tick **log** for a logarithmic scale and **bar** to show the legend in the view. Values that are not a number (NaN, fill values of NetCDF files) are drawn grey.

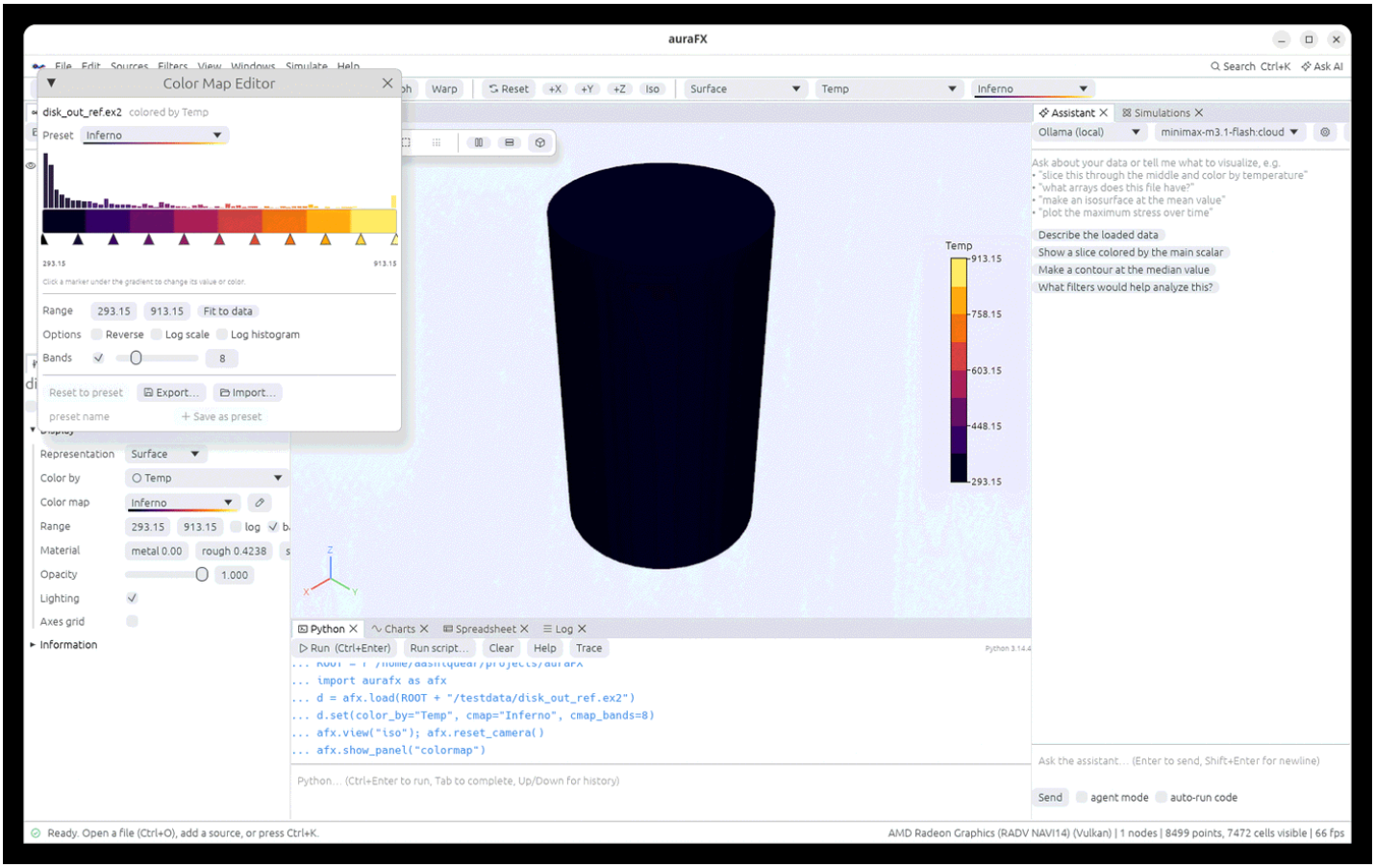
```
d.set(color_by="Temp", cmap="Inferno", range=(300, 900), log_scale=False)
d.set(color_by="V", component=0)        # x component of a vector
d.set(color_by=None, solid_color=(0.8, 0.2, 0.1))
```

To show an image on a surface instead of colors (a map on a globe), see Textures.

## Color maps

Eighteen maps ship with the app (Viridis, Inferno, Magma, Plasma, Cool to Warm, Jet, Rainbow, Grayscale, ColorBrewer sets and more); the **Color map** box shows each with a preview and marks the ones safe for color-blind readers. **Reverse** flips a map and **Bands** turns it into steps.

# The color map editor




**View › Color Map Editor…** (or the pencil next to the map) opens the editor for the selected object:

- the histogram of the array behind the map shows where the values lie;
- drag the markers to move colors, double-click to change a color, right-click to delete one, click between markers to add one;
- **Bands**, **Reverse**, and the range fields as in Properties;
- **Save as preset** keeps your map under a name for other objects and sessions;
- **Import…** reads ParaView color map files ( `.json` ) as presets; **Export…** writes them.

```
afx.colormaps()                                   # names
afx.define_colormap("hot steel", [[0, 0, 0, 0], [0.5, 1, 0.3, 0], [1, 1, 1, 1]])
afx.import_colormaps("paraview_maps.json")
d.set(cmap="hot steel", cmap_reverse=True, cmap_bands=8)
d.set(cmap_points=[[0, 0, 0, 1], [1, 1, 0, 0]])    # a one-off map on this object
```

# The legend

The legend (scalar bar) shows the map, its range and the array name for the selected object, or for the first object with a legend when nothing is selected. Untick **bar** to hide it. It is included in screenshots and animations, and drawn as real text in SVG exports.

# Views, layouts and the window

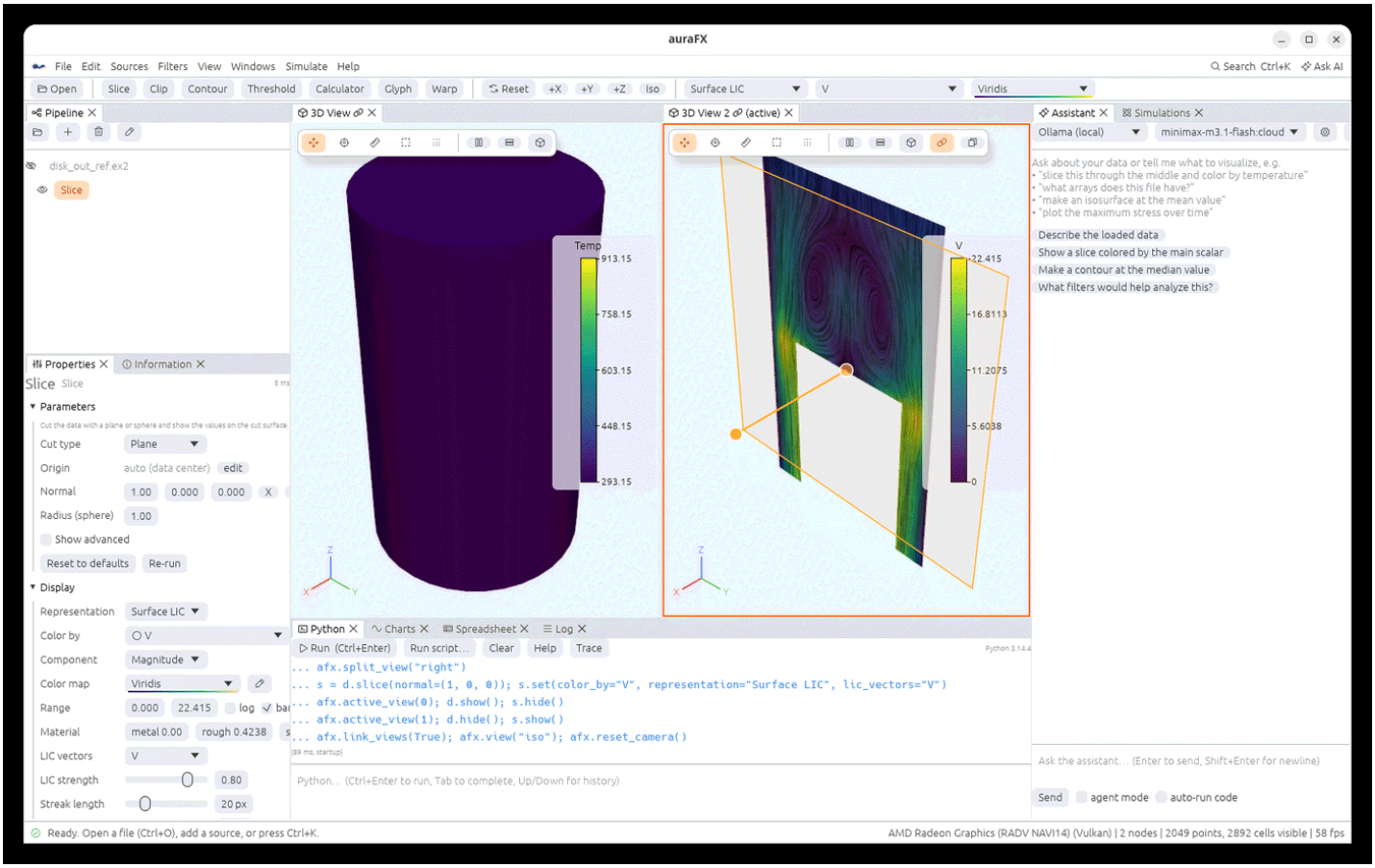
## The 3D view

| Action | Mouse / key |
|---|---|
| rotate | drag |
| pan | middle-drag, or Shift+drag |
| zoom | wheel, right-drag, or Ctrl+drag; pinch on a trackpad |
| fit the data | double-click, **R**, or the **Reset** button |
| look along an axis | **+X**, **+Y**, **+Z**, **Iso** in the toolbar, or **View › Look from…** |
| orthographic projection | **View › Orthographic projection** |
| orientation axes (bottom left) | **View › Orientation axes** |

The small toolbar inside the view switches the mouse mode: navigate (**Esc**), pick (**P**), ruler (**M**), select cells (**S**) and select points (**D**); see Selection, picking and widgets.

```
afx.view("iso"); afx.view("+x"); afx.view([1, 0.5, 0.3])   # a direction from the
data to the camera
afx.reset_camera()
```

# Several views




The split buttons in the view toolbar (or **Windows › Split view right / down**, `afx.split_view("right")`) add a 3D view next to the active one. Each view has its own camera and its own list of visible objects; click a view to make it active (orange frame), then the eye icons in the Pipeline and `node.show()` / `hide()` apply to it. From Python, `afx.active_view(1)` makes a view active (0 is the first one) and returns once it is, so the lines after it act on that view:

```
afx.split_view("right")                  # returns the new view's number (1)
afx.active_view(0); d.show(); s.hide()    # left: the data
afx.active_view(1); d.hide(); s.show()    # right: the slice
```

- **Link** (the chain button, `afx.link_views(True)`) keeps the cameras of the linked views together; **Match** (`afx.match_views()`) copies the active camera to the others once.
- Right-click an object › **Send to view** moves it to another view or into a new one (`node.send_to_view(1)` or `"right"`, `keep=True` to keep it in the current view too).
- Close a view with the × on its tab; its objects stay in the pipeline.

## Panels and layouts

The **Windows** menu lists every panel (Pipeline, Properties, Information, 3D views, Python, Charts, Spreadsheet, Log, Assistant, Simulations) with a tick to show or hide it, and four layout presets:

| Preset | Shows |
|---|---|
| **Visualize** | pipeline, properties and a large 3D view |
| **Analyze** | the view with charts and the spreadsheet |
| **Script** | the view with a tall Python console |
| **Simulate** | the simulations panel, the log and the view |

**Reset layout** returns to the default. Panels can be dragged by their tabs into new positions or into tabs of each other; the arrangement is remembered between sessions.

```
afx.layout("analyze")
afx.show_panel("Charts"); afx.hide_panel("Log")
```

## Theme and scale

**Settings › Theme** chooses dark, light or follow-the-system; the 3D background follows the theme (and can be set per theme). **UI scale** enlarges the whole interface. From Python: `afx.set_theme("light")`.

## Command palette

**Ctrl+K** (or Ctrl+Space, or the **Search** button) opens the palette: type a few letters of a filter, a source, a panel or an action (open, save screenshot, reset camera, undo, light theme, ask the AI…) and press Enter. It is the fastest way to reach anything in the app.

## Undo, redo and state

**Edit › Undo / Redo** (Ctrl+Z, Ctrl+Y) cover opening files, adding and deleting filters, parameter and display changes. **File › Save State** writes everything to an `.afx` file (see Getting started); `afx.save_state("work.afx")` and `afx.load_state("work.afx")` do the same from Python.

# Selection, picking and widgets

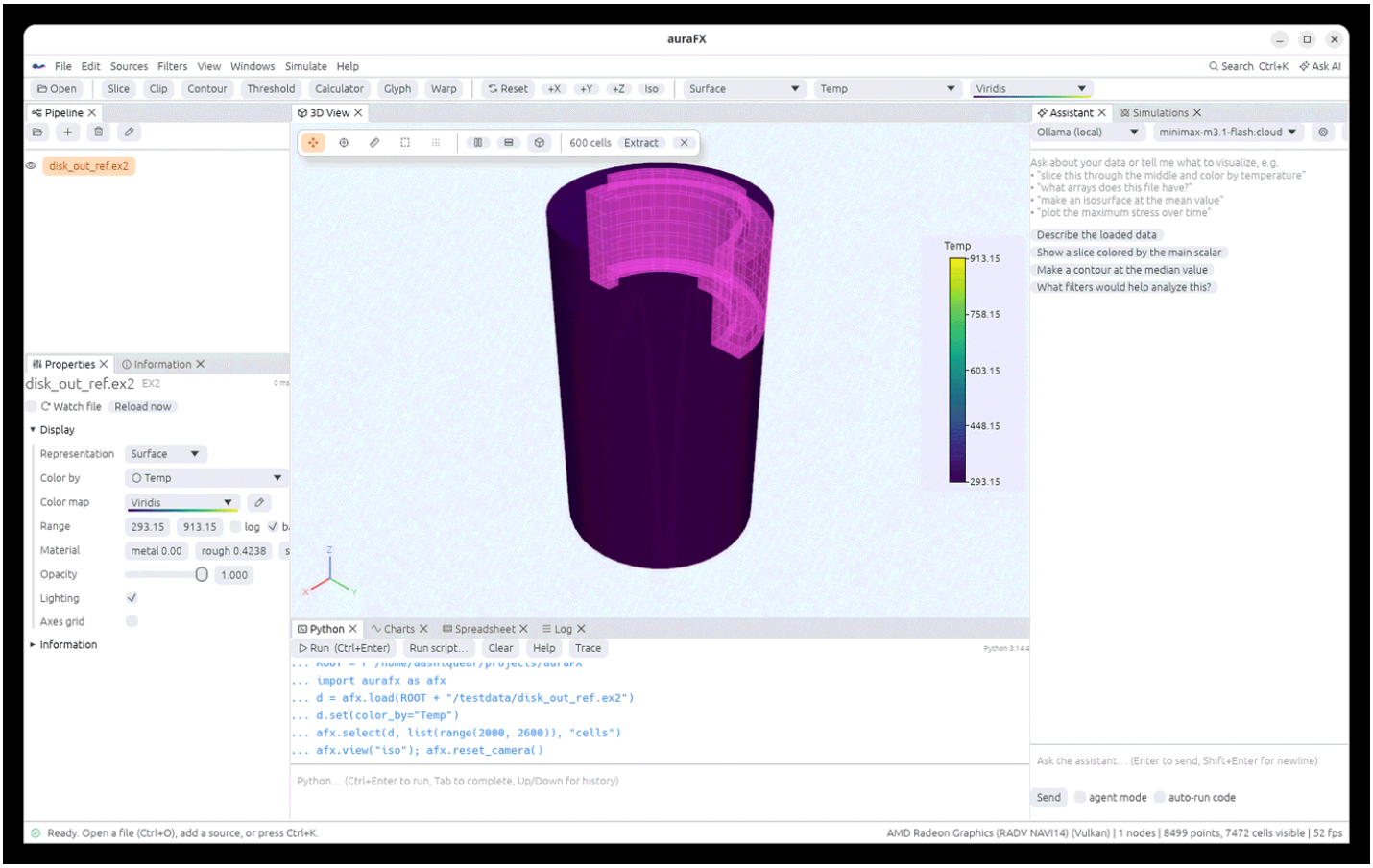
## Picking values

Press **P** (or choose the pick mode in the view toolbar, or Ctrl+click in any mode) and click a surface: the point is marked and the Information panel lists every array's value there, with the object's name. `afx.probe(node, (x, y, z))` returns the same values from Python.

## The ruler

Press **M**, click two points on surfaces: the distance is drawn and printed between them. Press M again (or Esc) to leave the mode.

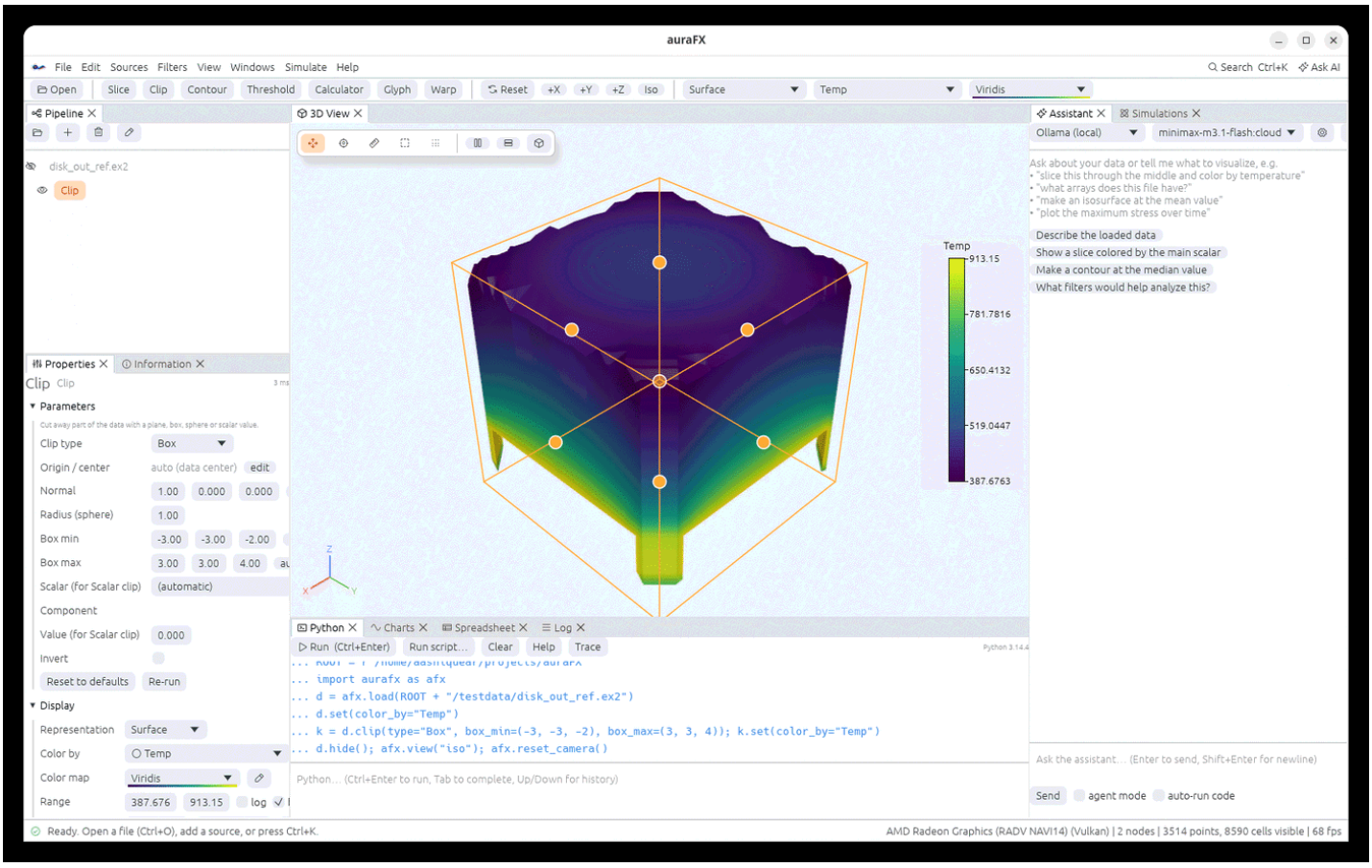
## Selecting cells and points



Press **S** (cells) or **D** (points) and drag a rectangle over the view. The selected cells are drawn highlighted through the surface. **Shift** adds to the selection, **Ctrl** removes from it. By default only the cells you can see are selected; **Select through** in the view toolbar takes every cell inside the rectangle, hidden ones included.

- **Create an Extract Selection node** (the button in the view toolbar, `afx.extract_selection()`) turns the selection into a new object you can filter further.
- **Find Data** (Filters) selects by a query such as `Temp > 600` or the N largest values.
- The Spreadsheet shows the selected rows; `afx.selection()` returns them, `afx.select(node, ids, "cells")` sets them, `afx.clear_selection()` clears.

## Widgets: dragging filter shapes in the view




When the selected filter has a shape, it is drawn in the view with orange handles:

| Filter | Widget |
|---|---|
| Slice, Clip (plane) | a plane: drag the center along the normal, drag the tip to tilt |

| Filter | Widget |
|---|---|
| Clip (box), Box source | a box: drag the center to move it, a face handle to resize that side |
| Slice/Clip (sphere), Stream Tracer point-cloud seeds, Sphere source | a sphere: center and radius handles |
| Plot Over Line, Stream Tracer line seeds, particle tracers, Line source | a line: either end, or the middle to move both |
| Probe Location | a point |

The parameters in Properties follow the drag, and typing values moves the widget. Custom lights can be aimed the same way; see Lighting.

## Annotations

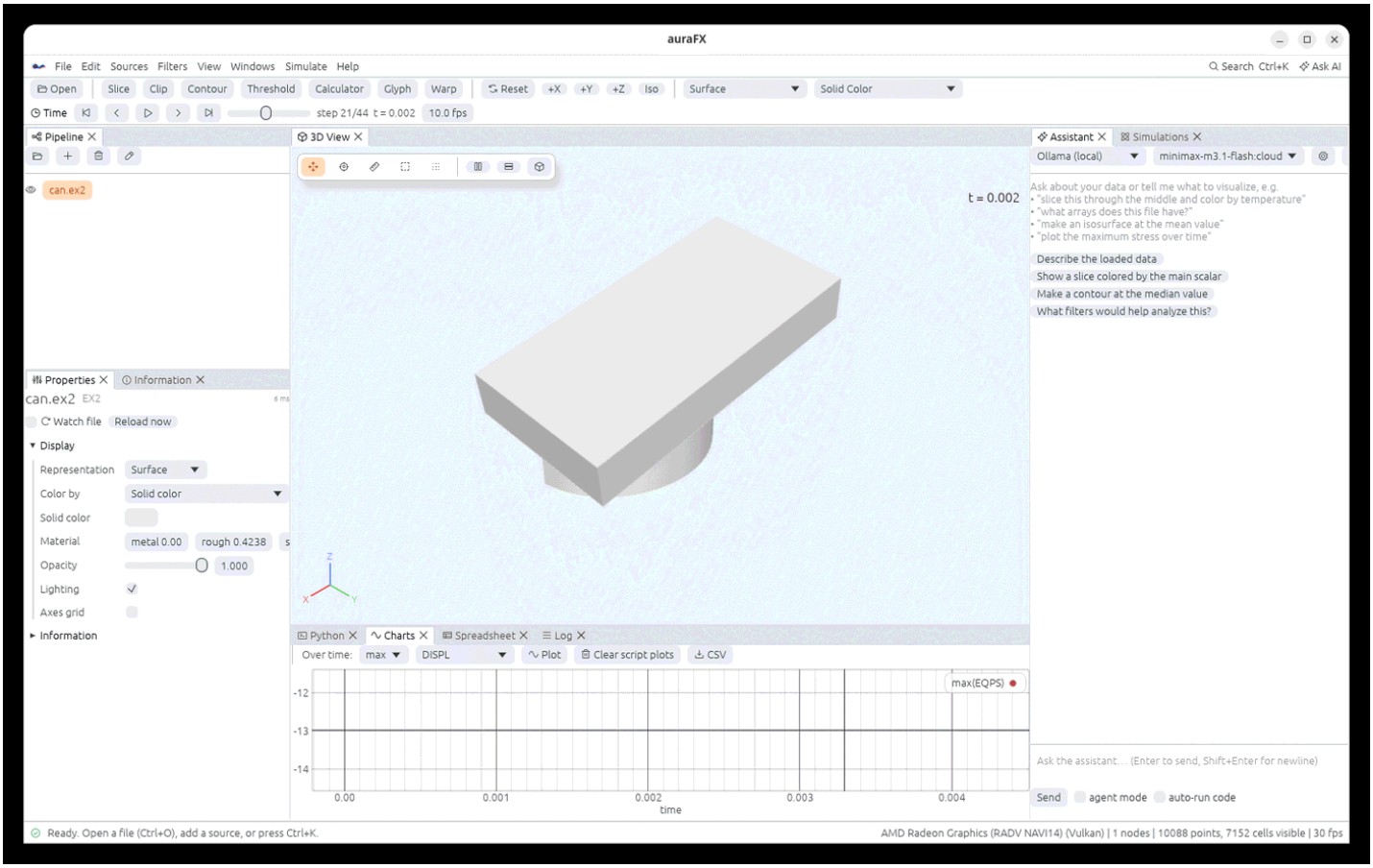
**Filters › Annotation**, `afx.annotate("Run 42", corner="upper left", size=18)` puts text in a corner of the view; `afx.time_label(True)` (or **View › Time label**) shows the current time. Both are drawn in screenshots and animations. `afx.clear_annotations()` removes them.

# Time, charts and animation

## Time steps

Files with several time steps (Exodus, `.pvd`, OpenFOAM cases, LAMMPS dumps, numbered file series, VTKHDF, EnSight, Xdmf, NetCDF, CGNS), and the Spatio Temporal Harmonics source, show the **time toolbar**: play/pause, first, previous, next, last, a slider, the step number, the time value and the playback speed. Every filter follows the time step; filters that look across time (Temporal Statistics, Particle Tracer, Temporal Interpolator…) read the steps they need.

```
afx.times()          # the time values
afx.time(20)         # go to step 20
```




## Charts

The **Charts** panel plots lines with zoom, legends and log axes:

- **Plot Over Line** (a filter) plots the arrays along a line;
- **Edit › Plot Over Time…** or `afx.over_time(node, "EQPS", stat="max")` plots a statistic (max, min, mean, sum, or the value of a global variable) over every time step;
- `afx.plot(x, y, label="T")` plots your own NumPy arrays; `afx.clear_plots()` empties the panel.

The **Spreadsheet** panel shows the selected object's points, cells or field data as a table with sorting, column choice and export to CSV.

## Animation

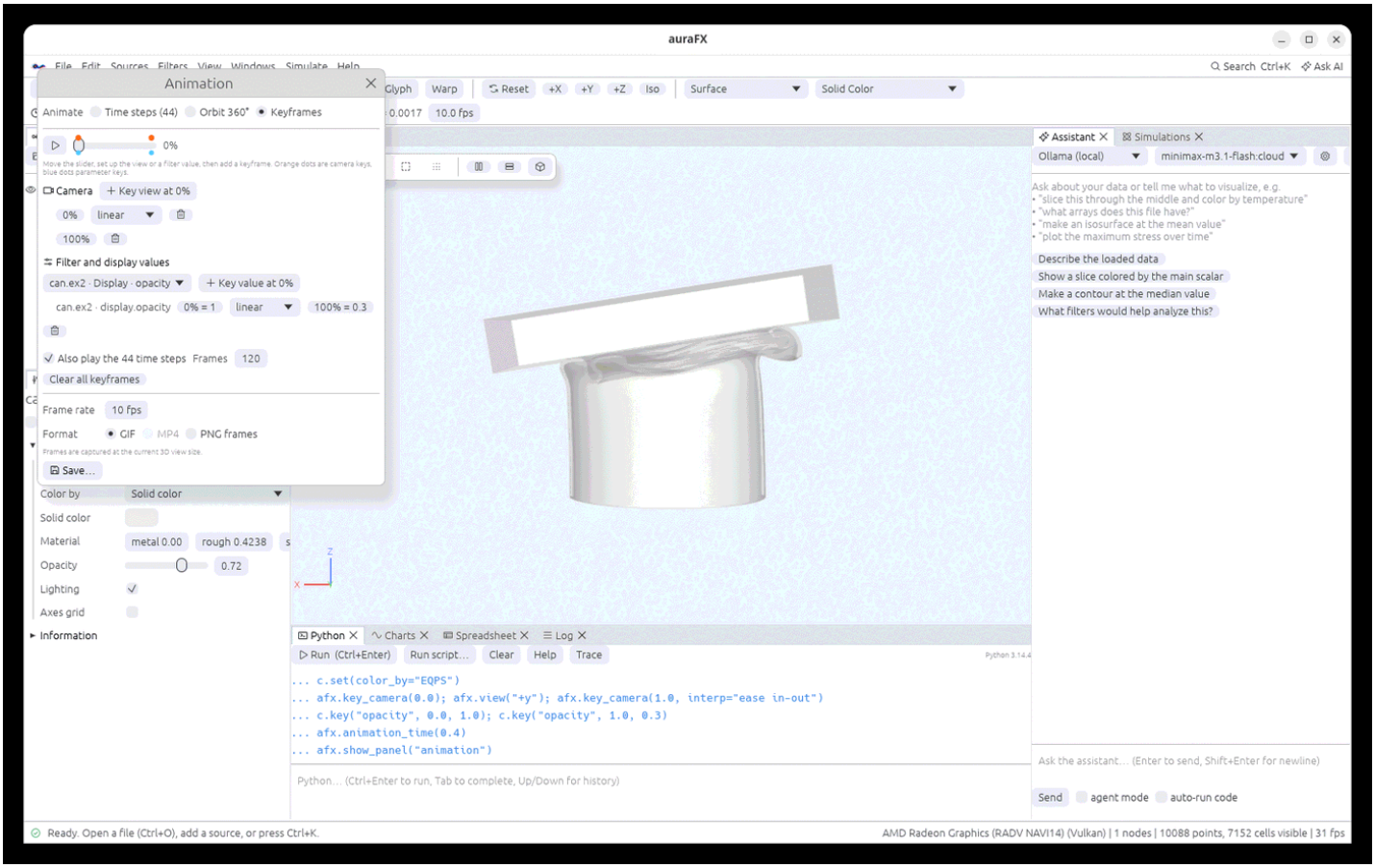
**File › Animation…** records one of three things:

| Mode | What it plays |
|---|---|
| **Time steps** | every time step of the data |
| **Orbit 360°** | the camera turning once around the data (choose the number of frames) |
| **Keyframes** | the keyframe timeline described below |

Pick the frame rate and the format: **GIF**, **MP4** (needs `ffmpeg` on the PATH) or a folder of **PNG frames**. Frames are captured at the 3D view's size, with the legend, labels and annotations.

```
afx.save_animation("run.gif", fps=10)          # time steps (or keyframes if any)
afx.save_animation("turn.mp4", orbit=72)       # a turntable
```

# Keyframes




The **Keyframes** tab of the Animation window is a timeline from 0 % to 100 %:

1. Move the slider, set up the view, press **Key view at …** to record the camera.
2. Select a filter or display value in the list (any numeric filter parameter, or opacity, point size, line width, colors…), set it, press **Key value at …**.
3. Each key has an interpolation menu for the way it moves on to the next key: **linear**, **step** (hold), **ease in**, **ease out**, **ease in-out**, **exponential** (constant ratio, for zooms and scales) or **wave** (there and back, a sinusoid).
4. Press play to preview; tick **Also play the time steps** to run the data's time along the animation; set the number of **Frames**; **Save…** records it.

The camera moves on a sphere around the focal point, so turns look even and zooms do not rush. Orange dots on the timeline are camera keys, blue dots value keys; click a key to jump to it, right-click to delete it.

```
afx.key_camera(0.0); afx.view("+y"); afx.key_camera(1.0, interp="ease in-out")
c.key("opacity", 0.0, 1.0); c.key("opacity", 1.0, 0.3)
s.key("origin", 0.5, (0, 2, 0), interp="wave")
afx.animation(frames=120, time=True)
afx.animation_time(0.4)                          # preview a moment
afx.save_animation("keys.mp4", fps=24)
afx.clear_keyframes()
```

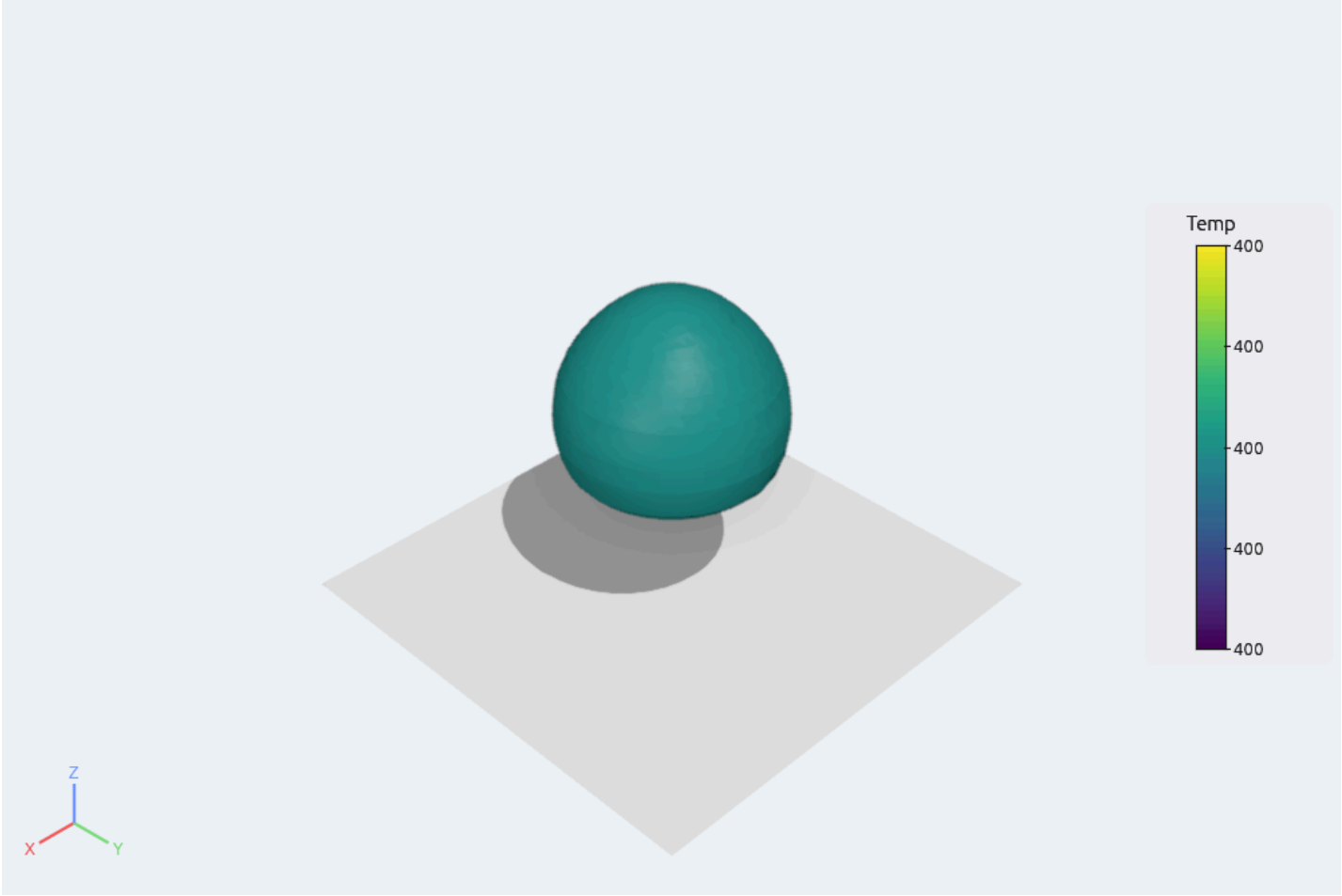
## Animations driven by a function

`afx.save_animation("a.mp4", cue=f, frames=60)` calls your function `f(t)` with the animation time `t` from 0 to 1 before each frame (after the keyframes are applied), so anything can change between frames: filter values, colors, the data loaded. The call returns when the file is written.

```
def cue(t):
    c.set(values=[350 + 500 * t])
afx.save_animation("contour_sweep.gif", fps=8, cue=cue, frames=40)
```

# Lighting, shadows and effects

Everything here is in **View › Lighting**, and in Python through `afx.lighting(...)`. Calling `afx.lighting()` without arguments returns the current defaults; unknown options raise an error, so typos are caught.




## Lights

- **Light kit** (on by default): a key light from the upper right, a fill light from the lower left and a head light, like ParaView's. Untick it for a single head light.
- **Custom lights**: tick it to edit the lights yourself. Each light has a color, an intensity, an azimuth and an elevation, follows the camera or stays fixed in the scene, and can cast shadows. **Add light** adds one; **ambient** sets the light from everywhere.
- **Show lights in the view** draws each custom light as a small sun around the data; drag a sun to aim its light.

```
afx.lighting(light_kit=False)
afx.lighting(lights=[{"direction": (0.5, 0.5, 1), "intensity": 0.8},
                     {"direction": (-0.7, -0.2, 0.5), "intensity": 0.3, "color":
(0.6, 0.7, 1), "camera_relative": False}],
             ambient=0.15)
```

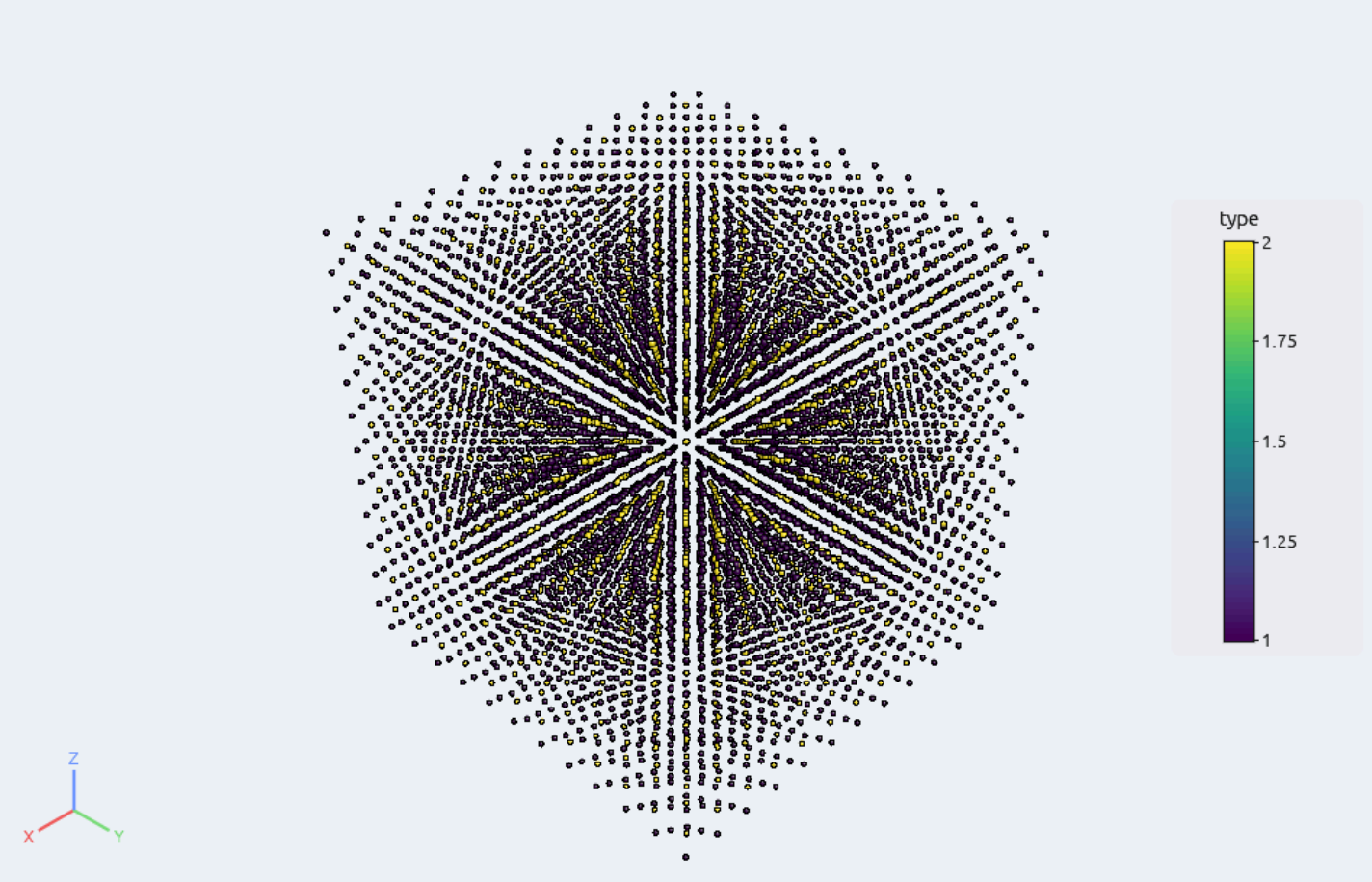
## Shadows

**Shadows** draws the shadow of the key light (or of the first custom light with shadows ticked) on every opaque surface: a shadow map fitted to the visible objects, softened at the edges. Objects drawn by a remote server are shadowed on that machine, so they do not cast shadows on local objects.

## Ambient occlusion

**Ambient occlusion** darkens creases, cavities and the contact between objects, which makes depth easier to read. The sliders set its strength and radius (a fraction of the data size). It works from the picture’s depth, so it costs the same whatever the mesh size.

# Eye-Dome Lighting




**Eye-Dome Lighting** shades every pixel by how much it stands out from its neighbours in depth: outlines and depth cues for point clouds, particles and dense meshes without any normals. The slider sets the strength. `afx.lighting(edl=True)`.

# Anti-aliasing, tone mapping and exposure

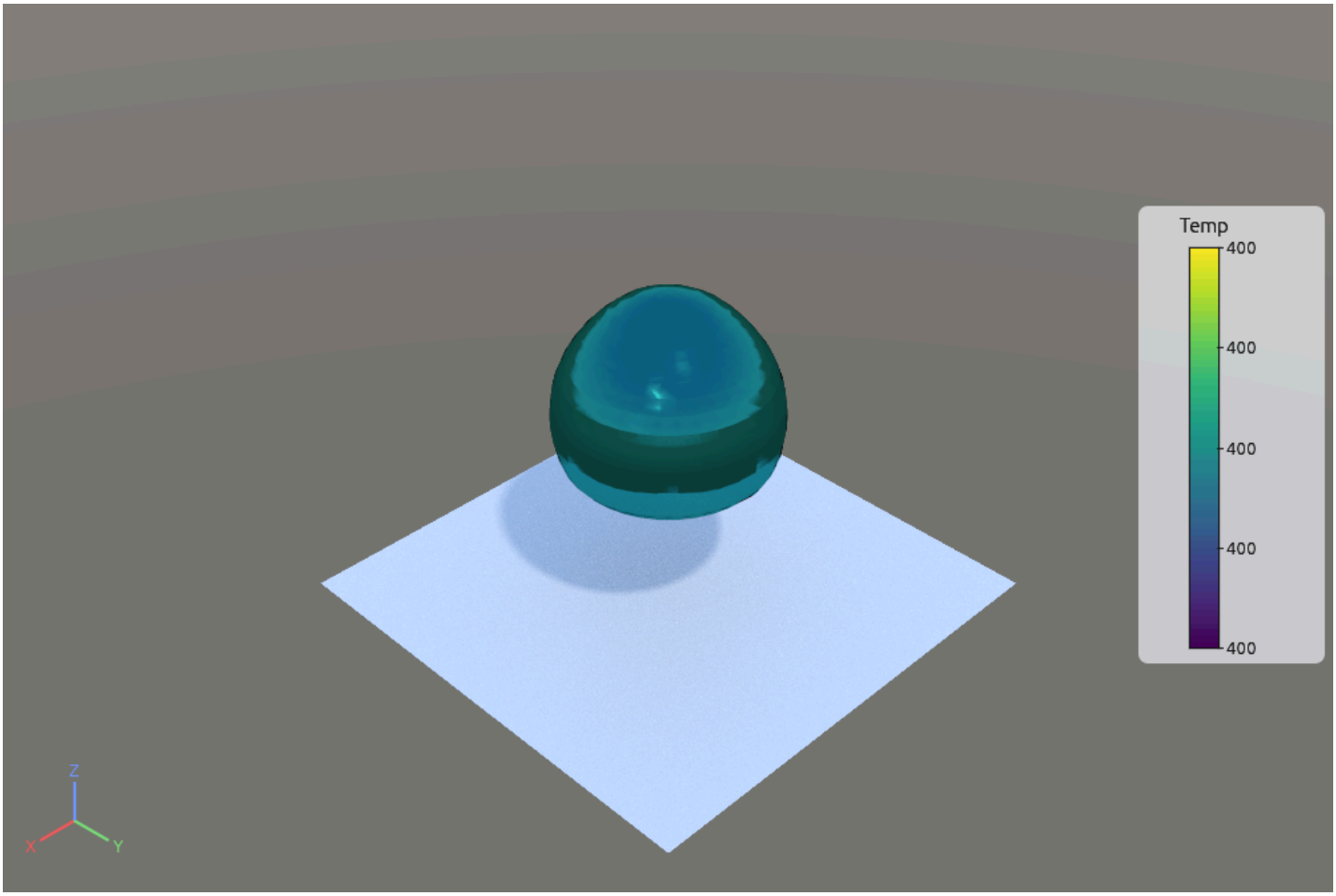
Edges are smoothed by multisampling (4 samples). **FXAA** adds a cheap post pass that smooths the remaining jagged edges of thin lines. **Tone mapping** compresses bright pictures with the **Reinhard** or **ACES filmic** curves, and **exposure** scales brightness before it.

```
afx.lighting(fxaa=True, tone_mapping="aces", exposure=1.2)
```

## Transparency

Translucent objects (**Opacity** below 1 in Properties) are blended in a way that does not depend on the order they are drawn in, so several overlapping translucent surfaces look right from any angle. Translucent objects are always drawn on this computer, even when the rest of the data is drawn by a server.

## Materials and environment maps




**Material** in Properties gives each surface a **metal** factor (0 plastic-like, 1 metal: reflections take the object's color), a **rough**ness (small: sharp highlights and mirror-like reflections; 1: matte) and a highlight strength. The defaults reproduce the plain shading.

**View › Lighting › Environment** lights the scene with a picture of the surroundings: the built-in **sky**, **studio** and **white** environments, or your own equirectangular `.hdr` / `.png` / `.jpg` (z up). It replaces the flat ambient light with soft directional light, adds reflections on shiny and metallic surfaces, and **as background** shows it behind the data. While an environment is on, the built-in lights are dimmed by half.

```
c.set(metallic=1.0, roughness=0.25, specular=0.5)
afx.lighting(environment="sky", environment_intensity=1.2, show_environment=True)
afx.lighting(environment="/data/hdri/studio_small.hdr")
```

## Ray tracing

**View › Lighting › Ray tracing** path-traces the opaque surfaces: soft shadows, light from the sky or environment, and light bouncing between surfaces. It refines while the camera is still (the rasterizer draws while you move) and the status corner shows the samples done. Set the number of **samples** per pixel to stop at, the number of **bounces**, and where it runs: **auto** (the GPU when the data fits in its memory, else the CPU), **GPU** or **CPU**. Edges, points, lines, translucent objects, volumes and the legend are still drawn on top. Screenshots wait for the tracer to finish and are traced at the requested size.

```
afx.lighting(ray_tracing=True, ray_samples=256, ray_bounces=3, ray_device="auto")
afx.screenshot("traced.png", width=3840)
```

# Surface LIC




The **Surface LIC** representation (Properties › Representation) draws a vector field as streaks on any surface while the surface stays colored by any array: the classic way to see a flow on a slice. Choose the vector array (**LIC vectors**), the streak **strength** and the **streak length** in pixels.

```
s.set(representation="Surface LIC", lic_vectors="V", color_by="Temp",
lic_length=30)
```

# Volumes, point clouds and curved cells

## Representations

**Representation** in Properties (and the toolbar) chooses how an object is drawn:

| Representation | Use |
|---|---|
| **Surface** | the outer surface, smooth-shaded (hard edges stay sharp: a 30° crease angle) |
| **Surface With Edges** | the surface with cell edges (**Edge color** in Properties) |
| **Wireframe** | edges only |
| **Points** | the points, as flat dots or shaded spheres (**Point size**, **spheres**) |
| **Outline** | the bounding box |
| **Volume** | GPU ray casting of image data (other meshes are resampled first) |
| **Surface LIC** | a vector field as streaks (see Lighting) |

**Lighting** can be switched off per object (flat colors, for 2-D data). **Axes grid** draws a labelled grid around the object with tick values.

## Volume rendering



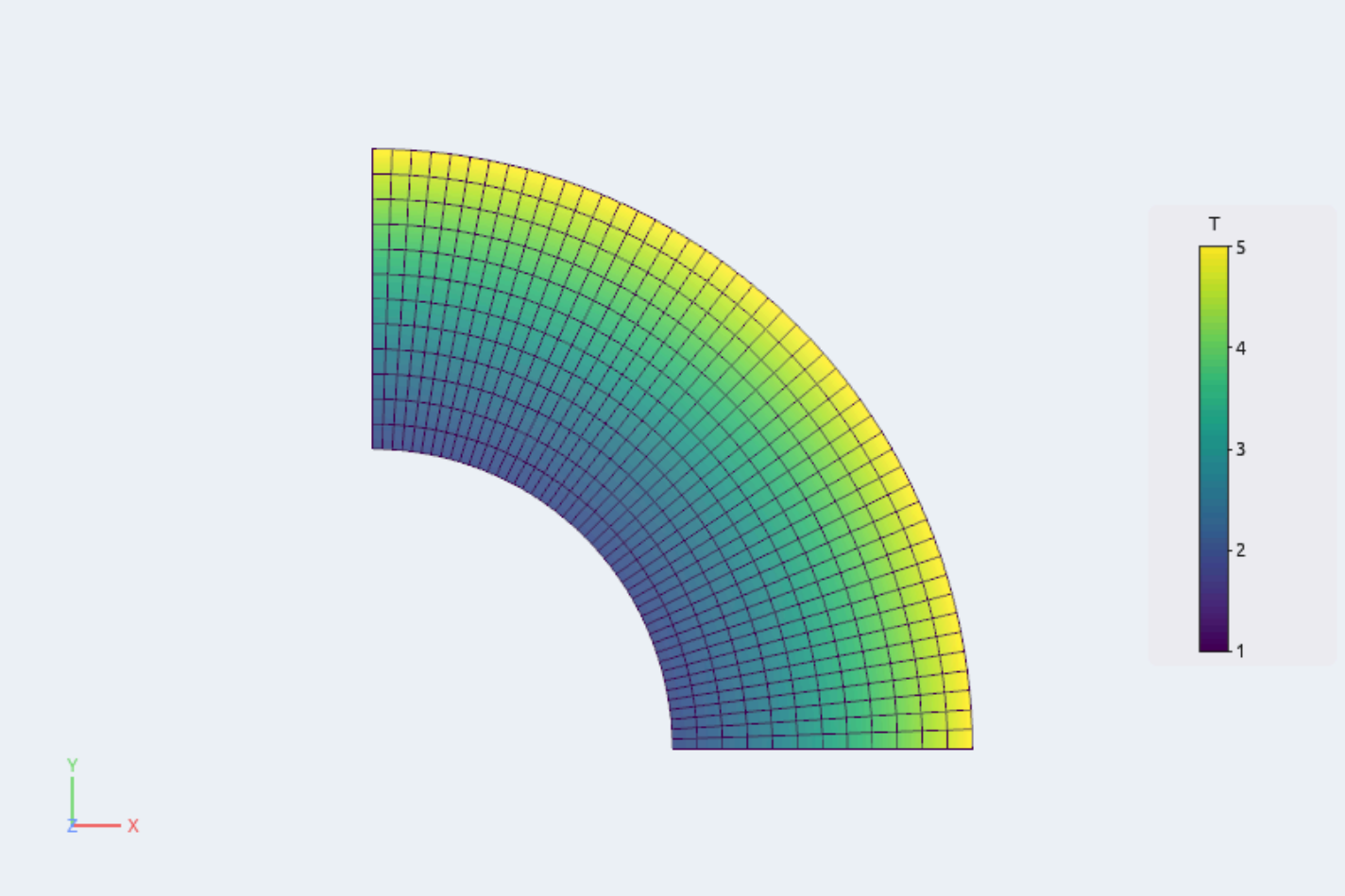

The **Volume** representation draws image data as a translucent cloud colored by the array: **Volume density** sets how opaque it is and **Transparent below** makes values under a fraction of the range invisible, which clears away the noise around a feature. Volumes are depth-merged with the surfaces around them.

```
w.set(representation="Volume", color_by="RTData", volume_opacity=2.0,
volume_threshold=0.3)
```

## Point clouds and particles

LAMMPS dumps, XYZ files, PDB files and point sources open as points. Draw them as spheres sized by **Point size**, color them by type, and switch on Eye-Dome Lighting to see the structure of a dense cloud. The **Glyph** filter draws arrows or spheres scaled by an array; **Tube** gives lines a thickness.

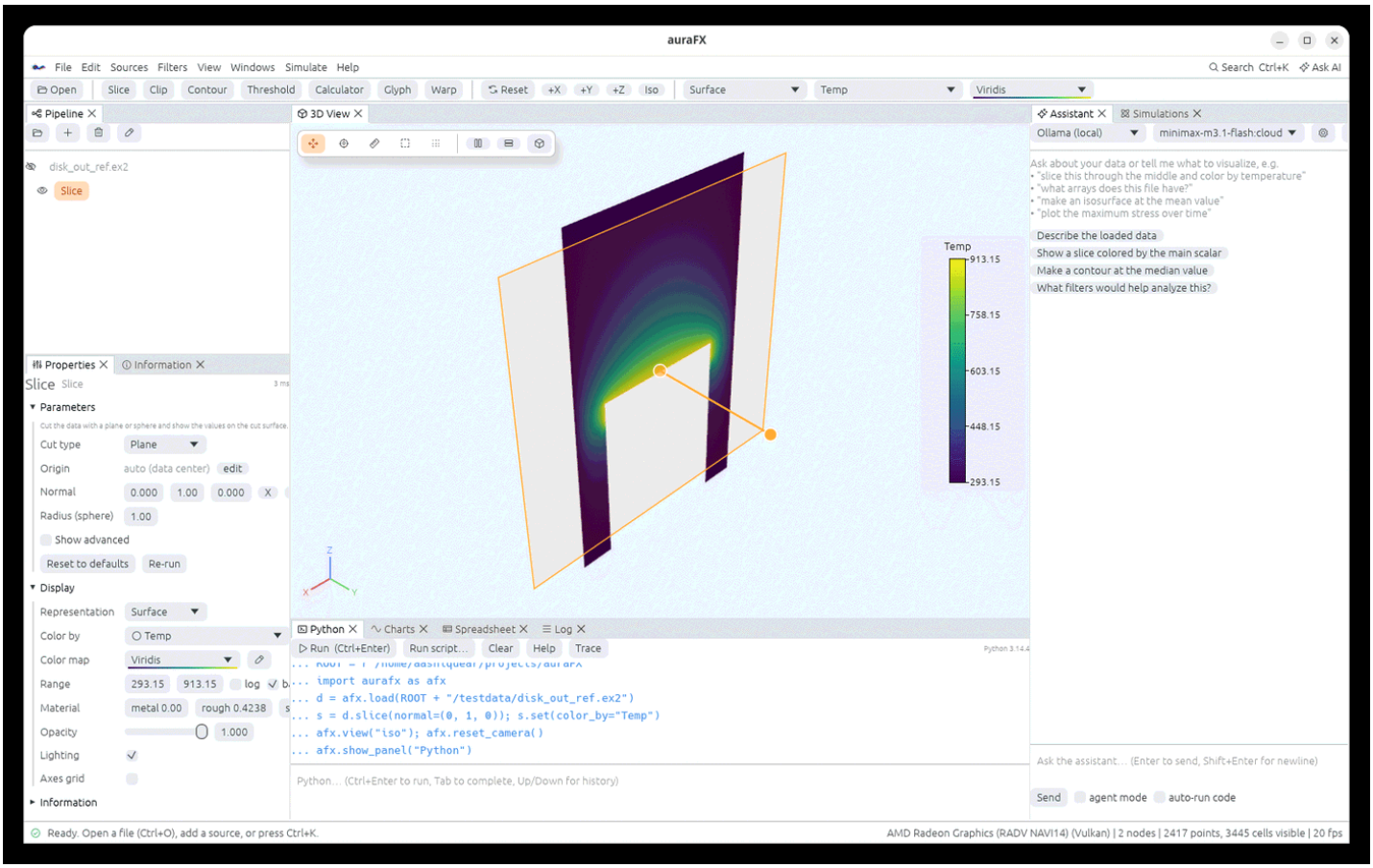
# Curved cells




Quadratic cells and Lagrange or Bezier cells of any order (VTK types 68–80, with rational weights and per-cell degrees) are drawn curved. **Curved cells › subdivision** in Properties sets how finely: 0 draws straight edges between corners, 1 passes through the nodes, each further level halves the pieces. Slice, Clip, Contour, Probe and Integrate Variables follow the curved shape, as ParaView does; the **Tessellate** filter turns curved cells into straight ones for filters that need them. The Information panel lists the cell types.

```
a.set(nonlinear_subdivision=3)
t = a.tessellate(level=2)
```

# Python




auraFX embeds Python with NumPy. The **Python** panel is a console: type a line and press **Ctrl+Enter** (or **Run**), use Tab to complete, Up/Down for history, **Run script…** to run a file, **Clear** to empty the log, **Help** for the cheat sheet. **Trace** turns what you did with the mouse into the equivalent code, which you can paste into a script. Everything the UI does is reachable from the `afx` module, and `aurafx --script file.py` runs a script at startup (the app stays open afterwards).

Packages installed into the embedded interpreter with `pip` (SciPy, pandas, PyTorch…) are available in the console.

## Loading and the pipeline

```
d = afx.load("run.e")                       # a Node; afx.load("slices/img_*.png")
stacks images
d = afx.load("/scratch/run.e", host="cluster")   # read on another machine (see
Data on another machine)
afx.nodes(); afx.active(); afx.get("Slice")      # find nodes
s = d.slice(normal=(0, 1, 0))               # any filter: node.<filter>(**params)
s.set(origin=(0, 1, 0), opacity=0.5)        # parameters and display settings
s.inputs(); s.activate(); s.delete()
s.update(); s.output()                      # wait; the data (an afx.Data: points,
arrays as NumPy)
s.info()                                    # arrays, ranges, bounds, counts
d.array("Temp")                             # a NumPy array; d.add_array("T_C", T -
273.15)
afx.apply_multi("append_datasets", [a, b])
afx.points_from(xyz, temperature=t)         # a node from NumPy
afx.filters(); afx.help_filter("contour")
afx.run()                                   # run whatever is pending
afx.undo(); afx.redo()
afx.save_state("work.afx"); afx.load_state("work.afx")
```

## Display

```
d.set(color_by="Temp", cmap="Inferno", range=(300, 900), log_scale=True,
representation="Surface With Edges")
d.set(opacity=0.6, point_size=6, points_as_spheres=True, line_width=2,
axes_grid=True)
d.set(metallic=1, roughness=0.2, specular=0.5)           # material
d.set(representation="Volume", volume_opacity=2, volume_threshold=0.3)
d.set(representation="Surface LIC", lic_vectors="V")
d.set(nonlinear_subdivision=2)                           # curved cells
d.show(); d.hide(); d.set_name("inlet")
```

## Views and camera

```
afx.view("iso"); afx.view("+x"); afx.view([1, 1, 0.5]); afx.reset_camera()
afx.split_view("right"); afx.active_view(1); d.send_to_view(1, keep=True)
afx.link_views(True); afx.match_views()
afx.layout("analyze"); afx.show_panel("Charts"); afx.hide_panel("Log");
afx.set_theme("light")
afx.annotate("Run 42", corner="upper left", size=18); afx.time_label(True);
afx.clear_annotations()
afx.lighting(shadows=True, ssao=True, edl=False, fxaa=True, tone_mapping="aces",
             environment="sky", ray_tracing=False)      # see Lighting
```

## Time, charts, selection

```
afx.times(); afx.time(10)
afx.over_time(d, "EQPS", stat="max")      # returns (times, values) and plots them
afx.plot(x, y, label="T", title="Temperature"); afx.clear_plots()
afx.temporal_statistics(d)
afx.select(d, ids, "cells"); afx.selection(); afx.extract_selection();
afx.clear_selection()
afx.probe(d, (0, 0, 1))                   # values at a point
```

## Pictures, animations, files

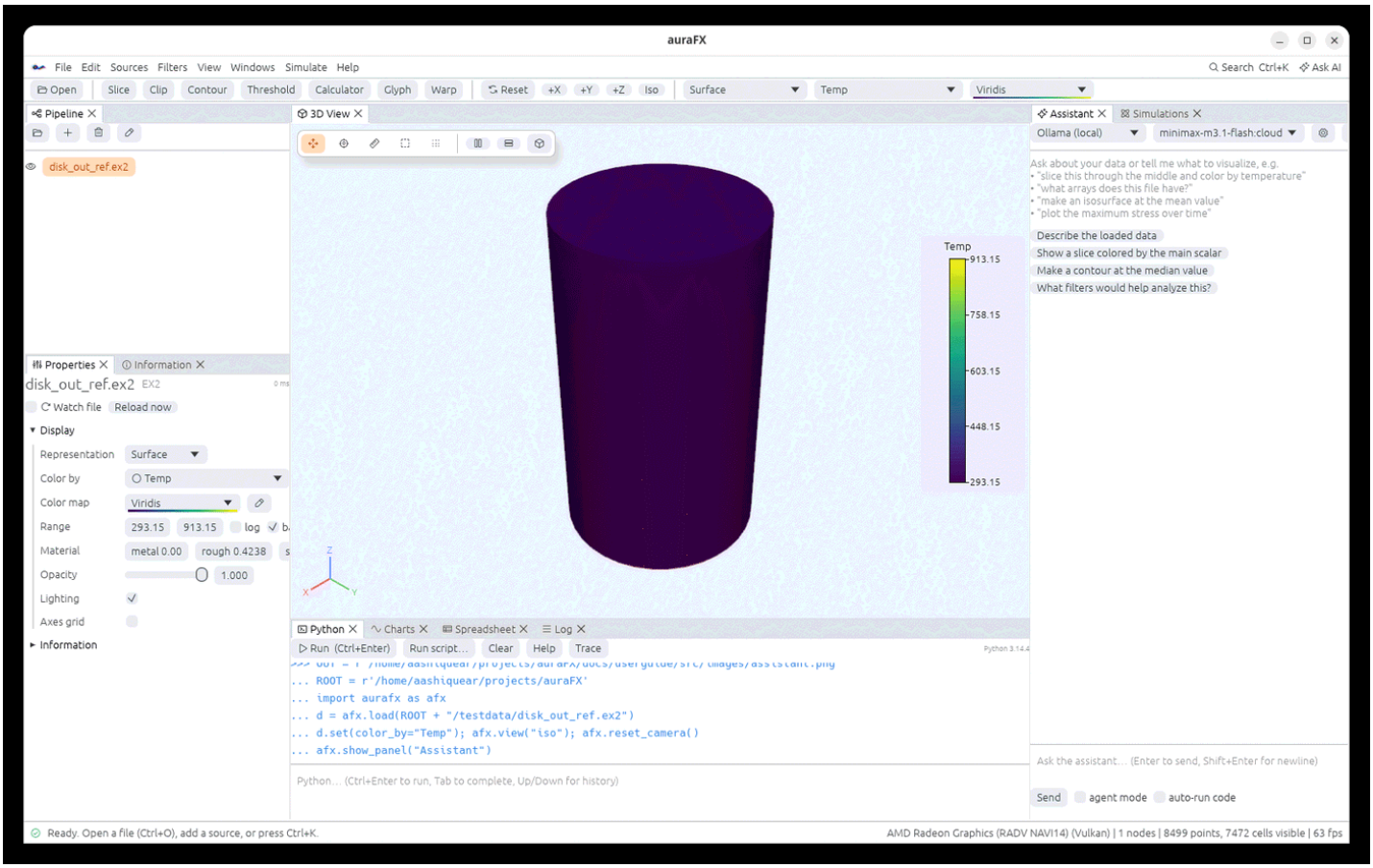
```
afx.screenshot("fig.png", width=3840, transparent=True, annotations=True)   #
waits until written
afx.screenshot("fig.svg")                 # vector picture
afx.export_scene("scene.glb")             # also .obj, .x3d, .wrl
afx.save_animation("run.mp4", fps=24); afx.save_animation("turn.gif", orbit=72)
afx.key_camera(0); afx.key_camera(1, interp="ease in-out"); d.key("opacity", 1,
0.2)
afx.animation(frames=120, time=True); afx.animation_time(0.5);
afx.clear_keyframes()
afx.save_animation("sweep.gif", cue=lambda t: c.set(values=[300 + 600 * t]),
frames=60)
d.save("out.vtu"); d.save("all.exo", all_times=True); d.save("all.vtkhdf",
all_times=True)
afx.colormaps(); afx.define_colormap(name, points);
afx.import_colormaps("maps.json")
```

## Assistant, plugins, simulations, servers

```
afx.ask("plot the maximum stress over time", agent=True)
afx.plugins(); afx.load_plugin("tools.py")
afx.submit("lmp -in in.melt", host="rocm-server", workdir="~/runs", gpu="amd:0",
outputs="dump.*", auto_load=True)
afx.sweep("lmp -var T {T} -in in.melt", T=[300, 600, 900], host="cluster")
afx.jobs(); afx.job(id); afx.cancel(id); afx.gpus("rocm-server"); afx.hosts()
afx.servers(); afx.load_remote("cluster", "/scratch/run.e")
```

The **Help › Python API cheat sheet** window lists the same calls inside the app, and every function has a docstring ( `help(afx.lighting)` ).

# AI assistant




The **Assistant** panel (right side; **Windows › Assistant**, or the **Ask AI** button) is a chat with a language model that knows the app and your data. Ask in plain words: “slice this through the middle and color by temperature”, “what arrays does this file have?”, “plot the maximum stress over time”, “why does my contour look cracked?”. The assistant answers and, when something should happen in the app, writes Python using the `afx` API.

## What the assistant sees

A text description of your pipeline: the objects, their filters and parameters, array names with ranges, bounds, cell types and time steps, plus the last error. It never receives the data itself. The description is rebuilt for every message.

## Running the code it writes

Code appears in a block with a **Run code** button; nothing runs until you press it, unless **auto-run code** is ticked. With **agent mode** on, the assistant runs its code itself, looks at the result (values, errors, the pipeline) and keeps going until the task is done, so “find where the plate yields and plot it” becomes several steps without you in between.

```
afx.ask("make an isosurface at the mean temperature")
afx.ask("find where the plate yields and plot the max stress over time",
agent=True)
```

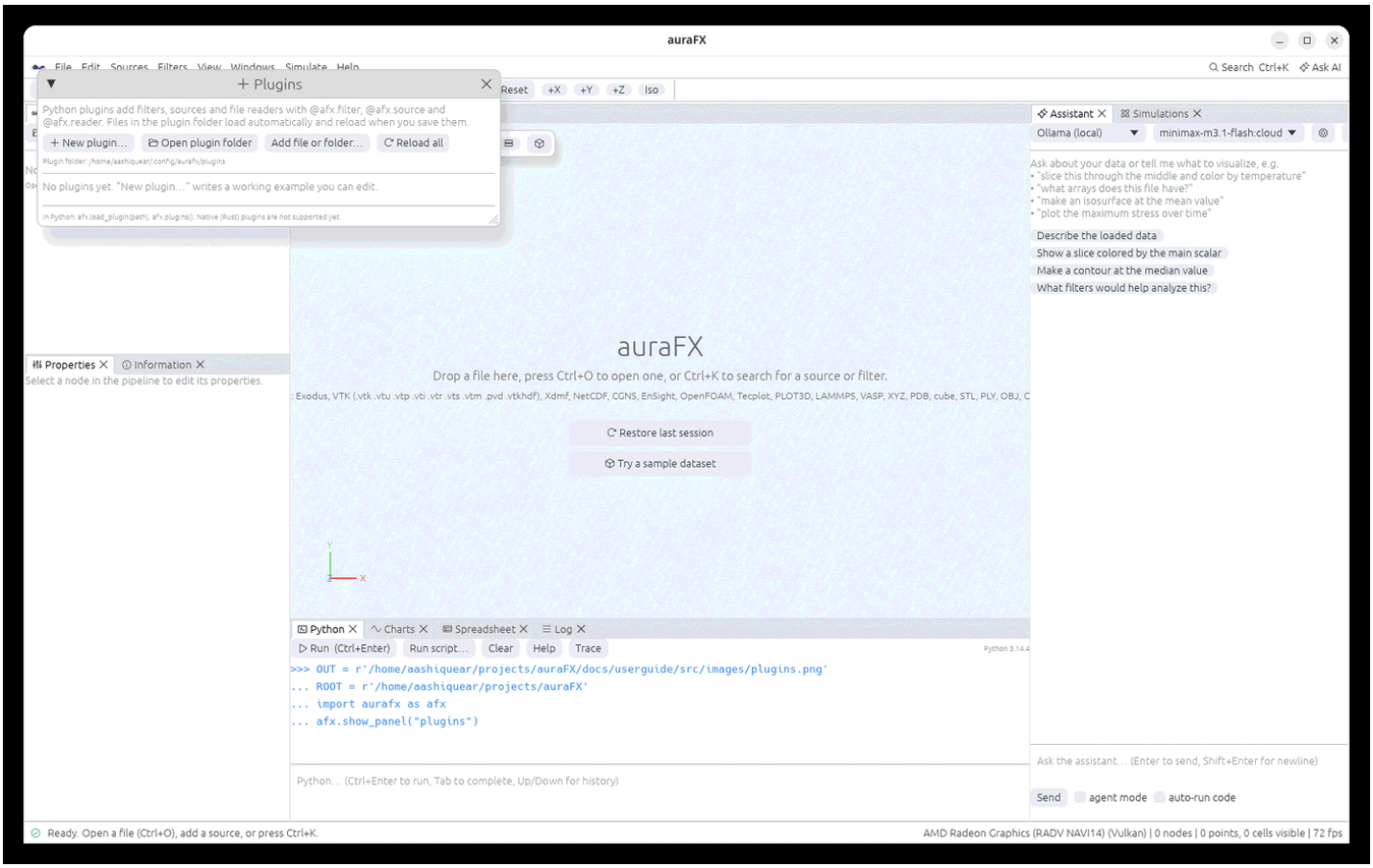
## Providers

Choose the provider and model in the panel’s drop-downs or in **Settings**:

| Provider | Setup |
|---|---|
| **Ollama** (local or cloud models) | the server URL from `OLLAMA_HOST` or Settings; the model list comes from the server |
| **Anthropic** | set `ANTHROPIC_API_KEY` or paste a key in Settings |
| **OpenAI-compatible** (OpenAI, OpenRouter, Groq, vLLM, LM Studio…) | the base URL and key in Settings |

Answers stream as they are produced. The suggestion buttons under the welcome text are good first questions for a new file.

# Plugins




A plugin is a Python file that adds filters, sources and file readers. They appear in the Filters and Sources menus, in the command palette, in Python ( `node.my_filter()` , `afx.my_source()` ), in the Open dialog (readers) and to the AI assistant, like the built-in ones.

## Where plugins live

- The plugin folder: **Edit › Plugins… › Open plugin folder** ( `~/.config/aurafx/plugins` on Linux, the auraFX config folder on macOS and Windows). Every `.py` file in it is loaded.
- Folders listed in the `AURAFX_PLUGIN_PATH` environment variable.
- Files or folders added with **Add file or folder…** in the Plugins window.
- From a script: `afx.load_plugin("path/to/tools.py")` .

Plugins reload when you save them, and the objects that use their filters re-run. Files whose name starts with `_` are skipped (use them for shared helpers). A plugin that fails to load shows its error in the Plugins window and the Log; switch a plugin off with its checkbox. **New plugin…** writes a working example and opens it in your editor.

## A filter

```
import aurafx as afx
import numpy as np

@afx.filter(category="Flow")
def speed(data, vectors=afx.ArrayParam(vector=True), name="speed"):
    """Magnitude of a vector array."""
    v = data.point_data[vectors]
    data.point_data[name] = np.linalg.norm(v, axis=1)
    return data
```

Keyword arguments become parameters in Properties; their type comes from the default: `float`, `int`, `bool`, `str`, a 3-tuple (a vector), `afx.Choice([...])`, `afx.ArrayParam(vector=False, cells=False)` (an array of the input) and `afx.Code("...")` (a text box). The docstring's first line is the description. Other decorator options: `label`, `description`, `keywords`, `id`, and `inputs=2` or `inputs=(2, None)` for filters with several inputs (the function receives a list).

`data` is an `afx.Data`, a copy of the input: `data.points` (N×3), `data.point_data["T"]`, `data.cell_data[...]`, `data.field_data[...]` (assign to add, `del` to remove), `data.n_points`, `data.n_cells`, `data.bounds`, `data.type`, `data.cells` (offsets, connectivity, VTK types), `data.blocks()`, `data.copy()`. ParaView spellings such as `data.PointData.append(a, "name")` work too. Build new data with `afx.make_points`, `afx.make_polydata`, `afx.make_image`, `afx.make_grid` and `afx.make_multiblock`.

## Sources and readers

```
@afx.source(category="Shapes")
def helix(turns=4.0, points=400):
    t = np.linspace(0, turns * 2 * np.pi, int(points))
    return afx.make_polydata(np.c_[np.cos(t), np.sin(t), t / 10], lines=
[list(range(len(t)))])

@afx.reader(extensions=["xyzt"], label="XYZ + temperature")
def read_xyzt(path):
    a = np.loadtxt(path, ndmin=2)
    d = afx.make_points(a[:, :3])
    d.point_data["T"] = a[:, 3]
    return d
```

Readers take precedence over the built-in readers for their extensions.

## Rules

- Plugin functions run on a worker thread while the pipeline runs: do not call pipeline functions (`afx.load`, `node.slice()`, `show`…) inside them.
- Ids must be unique and cannot replace built-in filters.
- For one-off code, the **Programmable Filter**, **Programmable Source** and **Python Calculator** run Python stored in the pipeline (and in state files).
- Native (Rust) plugins are not supported yet.

# File formats

Every reader was checked against ParaView 6.1.1: the same points, cells, bounds and value ranges on the same files. Where auraFX differs on purpose, the table says so.

## Readers

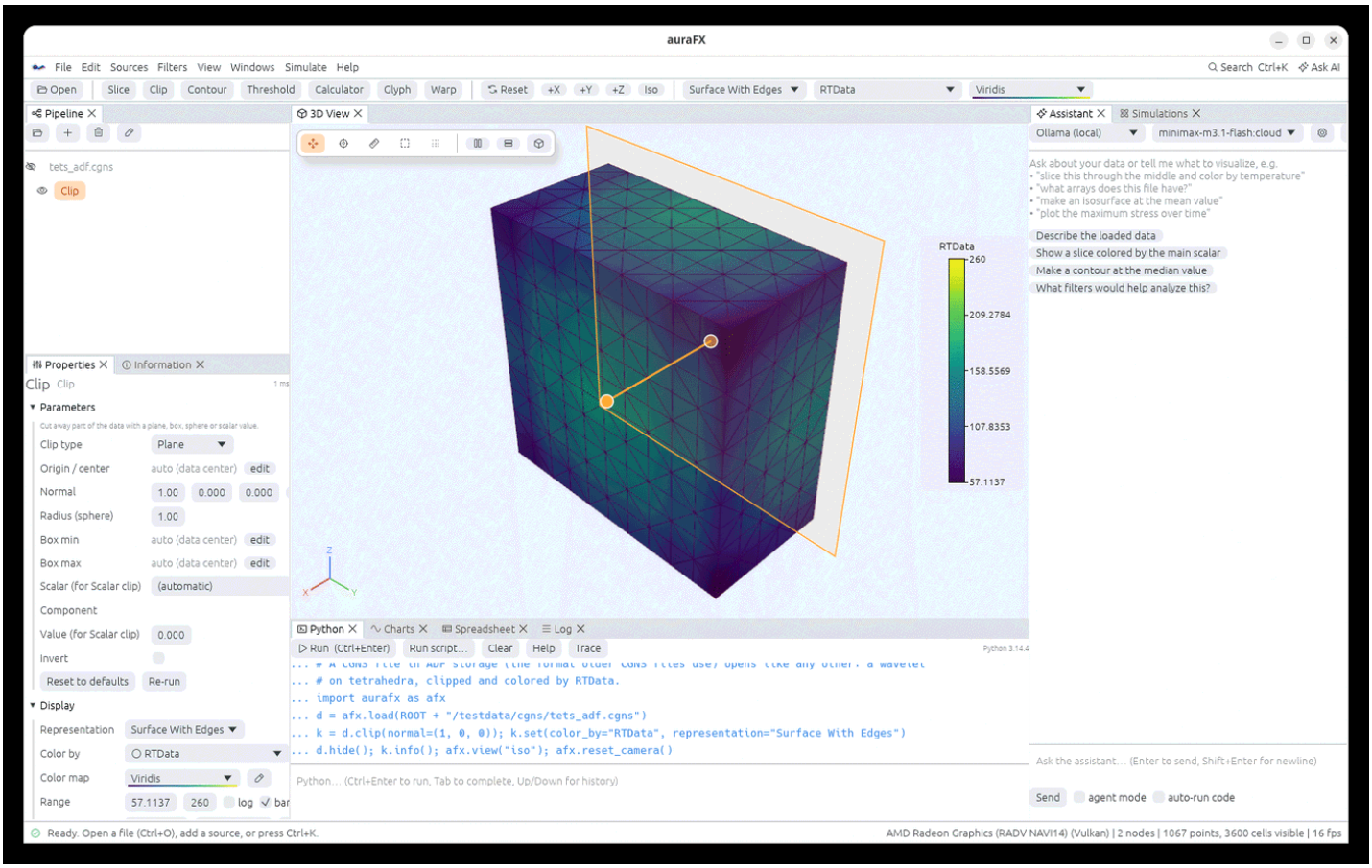
| Format | Extensions | Notes |
|---|---|---|
| Exodus II | `.e .exo .ex2 .g .gen` | NetCDF-3 and NetCDF-4; blocks, side and node sets, nodal/element/global variables, time steps, displacements |
| VTK legacy | `.vtk` | ASCII and binary, all dataset types |
| VTK XML | `.vtu .vtp .vti .vtr .vts .vtt`, parallel `.pvtu …`, `.vtm`, `.pvd` | inline, appended, base64; zlib and LZ4; curved (Lagrange/Bezier) cells |
| AMR (adaptive mesh refinement) | VTK `.vthb .vth`, AMReX/BoxLib plotfiles (the folder, or its `Header`) | levels of boxes with cell data; coarse cells under finer boxes are left out; see below |
| VTKHDF | `.vtkhdf .hdf` | image data, unstructured grids, poly data, partitioned collections with their assembly, time steps; chunked and gzip storage; no HDF5 library needed |
| OpenFOAM | `.foam` | ASCII/binary/gzip, decomposed and reconstructed cases, time directories, boundary patches |
| EnSight Gold | `.case .encas` | ASCII and C binary, all element types, structured blocks, time sets, wildcard file names |
| CGNS | `.cgns` | HDF5 and ADF storage; structured and unstructured zones, MIXED and NGON/NFACE sections, solutions, time steps |
| Xdmf 2/3 | `.xmf .xdmf` | inline, HDF5 and raw binary data; collections; `xi:include` and XPath |

| Format | Extensions | Notes |
|---|---|---|
| | | references (FEniCS/Dolfinx); HyperSlab and JOIN items |
| NetCDF (CF) | `.nc .nc4 .cdf` | classic and NetCDF-4; lon/lat grids on a sphere with cell bounds, curvilinear grids with 2-D longitude/latitude (POP, CICE, ROMS), rectilinear otherwise; time steps; packing; fill values become NaN (grey); see below |
| CAM-SE | `.nc` + connectivity file (found from the contents) | CESM atmosphere on its cubed sphere: lon/lat map or sphere, surface, midpoint or interface variables, one level or all; see below |
| MPAS | `.nc` (found from the file's contents) | MPAS ocean, atmosphere and land-ice output: cell centers joined into triangles, any vertical level or all of them, sphere or lon/lat map, time steps; see below |
| Tecplot | `.dat .plt .tec .tp` | ASCII (ordered and FE zones, BLOCK/POINT, cell-centered variables) and binary `.plt`; `.szplt` is not read |
| PLOT3D | `.xyz .x .p3d .bin` + `.q` | layout detected from the file (Fortran/C binary, byte order, precision, multi-grid, 2-D/3-D, IBLANK), ASCII |
| LAMMPS | `.dump .lammpstrj .atom` (+ `.gz`), `.data .lmp` | multi-frame dumps with any columns, triclinic boxes; data files with bonds |
| XYZ / extended XYZ | `.xyz .extxyz` | multi-frame; properties and lattice from the comment line |
| PDB | `.pdb .ent` | all atoms (hydrogens kept), MODELs as time steps, CONECT and distance bonds, residues and chains |
| Gaussian cube | `.cube` | the density as a volume in Å, plus the atoms with bonds |
| VASP | `POSCAR`, `CONTCAR`, `CHGCAR`, `LOCPOT`, `ELFCAR`, `PARCHG` (by name), `.vasp` | atoms with bonds and the lattice; volumes on the cell's axes (charge per Å³) |
| Images | `.png .jpg .bmp .tif` | as image data; multi-page TIFFs and `img_*.png` wildcards as volumes |

| Format | Extensions | Notes |
|---|---|---|
| GIS rasters | GeoTIFF `.tif`, Esri ASCII grid `.asc` | at their map coordinates, a cell per pixel, every band; no-data values become NaN (grey); see below |
| GIS vectors | Shapefile `.shp`, GeoJSON `.geojson` `.json`, KML `.kml`, GeoPackage `.gpkg` | points, lines and polygon outlines (or filled areas) with their attributes as cell data, a block per layer; see below |
| Surfaces | `.stl` `.ply` `.obj` | |
| CSV / TSV | `.csv` `.tsv` `.txt` | delimiter and header detection; tables with x, y, z columns become points |

Numbered files (`run_0001.vtu` …) open as a series when one of them is opened.

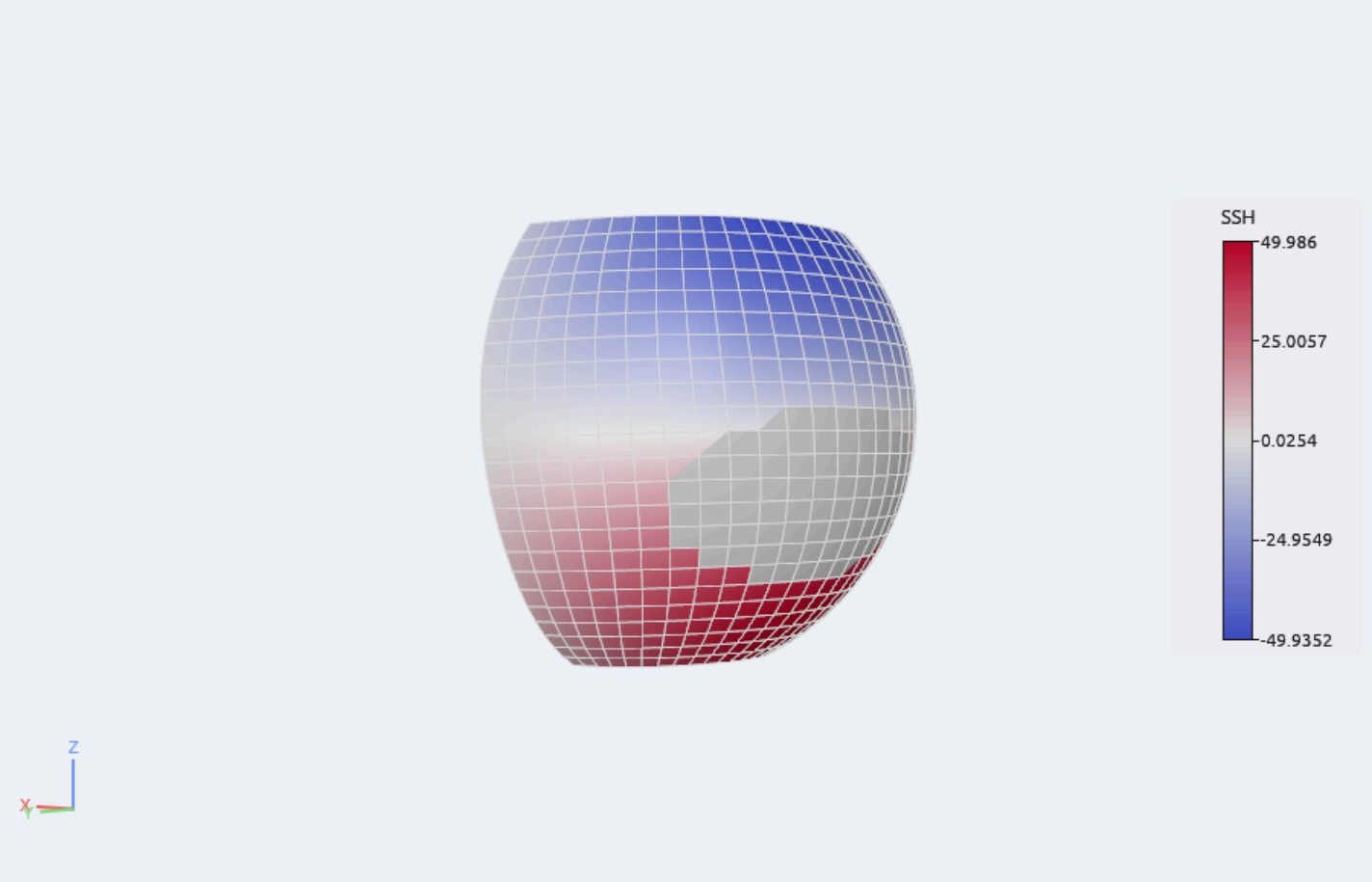
CGNS files come in two storage formats with the same `.cgns` extension: HDF5, the default since CGNS 3.4, and ADF, which older solvers and older CGNS libraries write. auraFX looks at the file and reads either one the same way, with **File › Open** or `afx.load("wing.cgns")`. Both the 64-bit ADF layout and the older 32-bit one are read, in either byte order. Links inside the file and links to other ADF files are followed. A link to an HDF5 file from an ADF file is not.



## NetCDF climate and ocean files

Files that follow the CF conventions are placed by their coordinates:

- Longitude and latitude axes put the grid on a sphere of radius 1. A third (vertical) axis becomes the radius, as in ParaView. Depths marked `positive="down"` are turned around, and all levels are shifted so that none is below zero, so deeper levels lie further in (the deepest at the center).
- Ocean and sea-ice models such as POP (CESM's ocean), CICE and ROMS use index dimensions (`nlat`, `nlon`) and give each point's longitude and latitude in 2-D variables (`TLONG`, `TLAT`) named in the `coordinates` attribute. These become curvilinear grids on the sphere. The coordinate variables themselves are not shown as data.
- When the variables live on different grids (say `TEMP` on depth, latitude and longitude, and `SSH` on the surface only), **Reader options › Variables on** picks the grid. Python: `ds.set(variables="(nlat, nlon)")`. The grid with the most dimensions is read first.

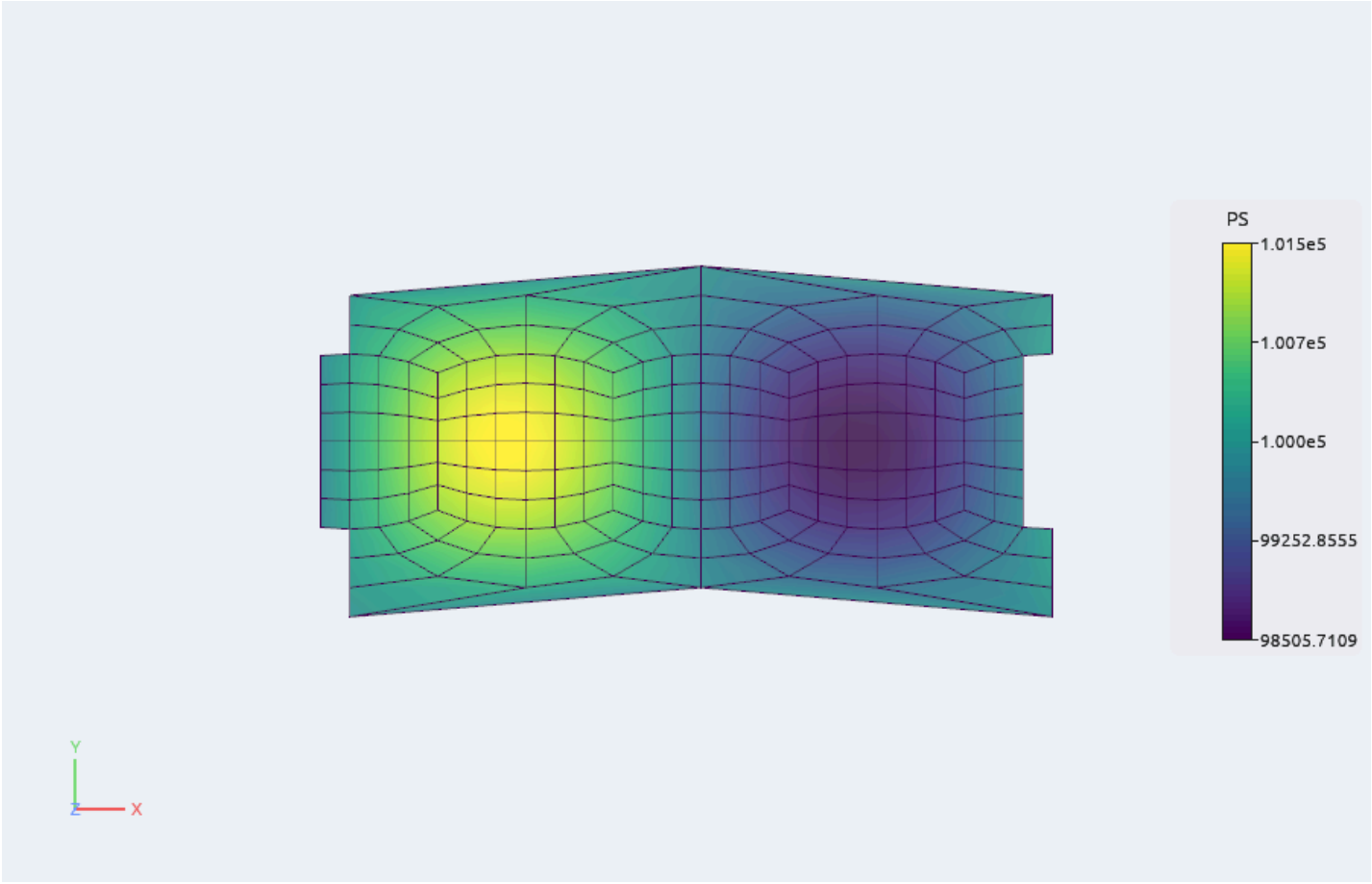



ParaView's dedicated POP reader only reads variables with exactly three dimensions, so it skips POP history output (time, depth, latitude, longitude). Its general NetCDF reader places those files as auraFX does.

CAM-SE output (the spectral-element core of CESM's atmosphere model) comes as two files: the data, with columns `ncol` at `lat` / `lon`, and a connectivity file whose `element_corners` joins

the columns into quads. Open the data file. auraFX finds the connectivity file in the same folder, or you can name it under **Reader options › Connectivity file** (Python: `ds.set(connectivity="ne30np4_connectivity.nc")`). Like ParaView's CAM reader, it draws the grid on a flat lon/lat map:

- Quads that cross the 0/360° seam are moved to the side where they stick out least, on copies of their points, so the map runs a little past 0 and 360.
- **Variables** picks the surface variables `[time, ncol]`, or those on midpoint (`lev`) or interface (`ilev`) levels. With **All levels** on, the default, each quad becomes a column of hexahedra, the top level highest, and the `lev` (or `ilev`) array holds each point's level. With it off, **Level** chooses one level.
- **Layout › Sphere** puts the columns on a sphere instead. This is not in ParaView; there is no seam on a sphere.
- A single level lies at height 0 (ParaView puts it at 1 or at the number of levels).




MPAS output (from the ocean, atmosphere or land-ice models of the Model for Prediction Across Scales) is NetCDF too. When a `.nc` file has MPAS's mesh variables (`xCell`, `cellsOnVertex`), auraFX reads it as MPAS instead of as a CF grid, the way ParaView's MPAS reader does:

- Each MPAS cell center becomes a point, and the cells around each MPAS vertex are joined into a triangle (or a quad, on quad meshes). On a sphere, the points lie at the planet's

radius in meters.
- Variables on cells ( `temperature` , `ssh` …) become point data, and variables on vertices ( `relativeVorticity` …) become cell data. Time steps are numbered 0, 1, 2…, and the date of each step ( `xtime` ) is kept with the data.
- For other dimensions than time and position, the first index is used, except the vertical levels, which you choose with the reader options below.

### Reader options

Some readers have settings for the file they read. They appear under **Properties › Reader options** when the file is selected, and from Python they are set like any other property, with `node.set(...)` . Changing one reads the file again; the choice is saved with the state file. **Reset to defaults** goes back to the file as first opened.

An MPAS file offers:

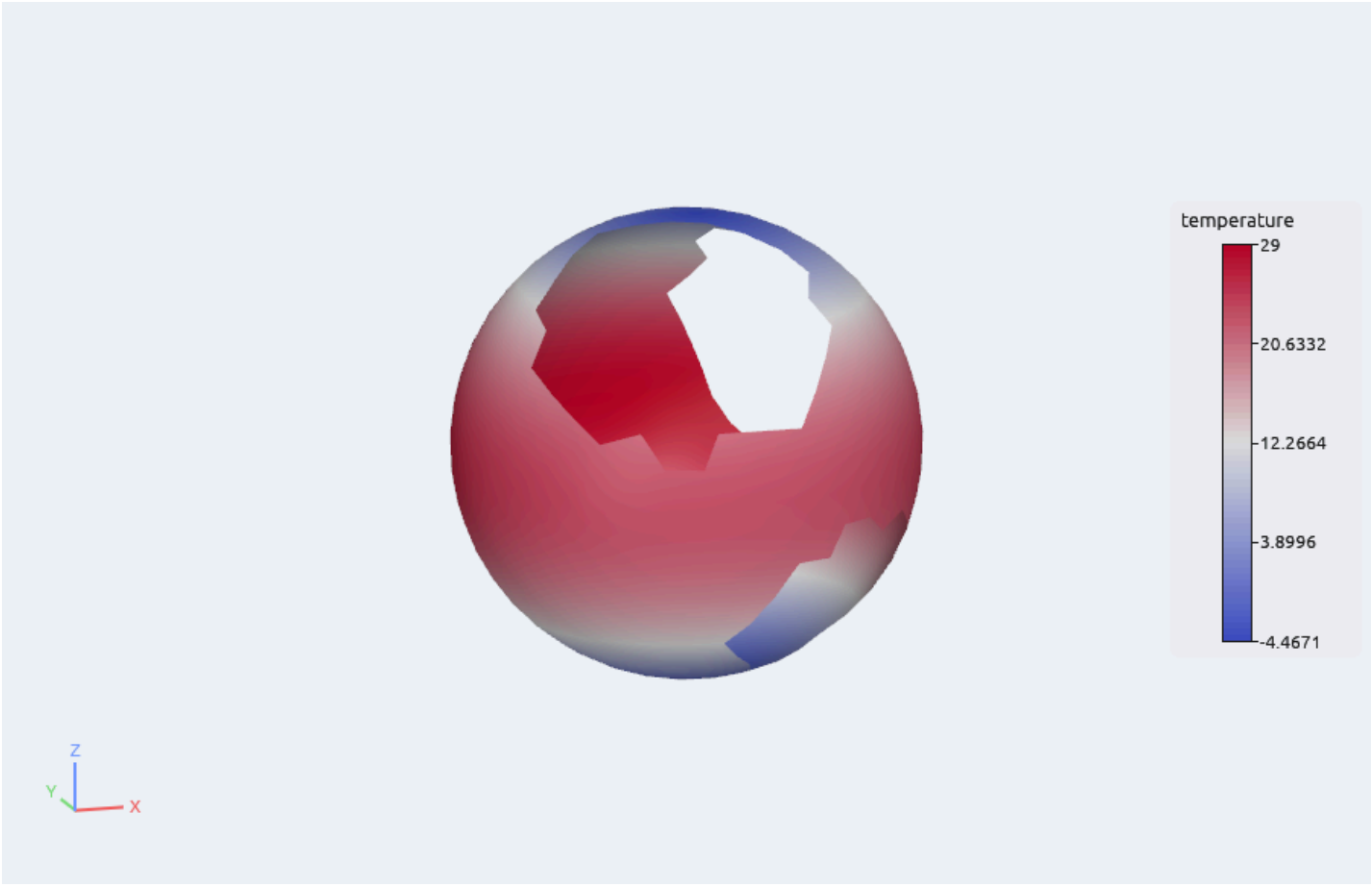
| Option | Python | What it does |
|---|---|---|
| Vertical level | `vertical_level=2` | which level of 3-D variables is shown (0 is the top of the ocean, the bottom of the atmosphere) |
| All levels (multilayer) | `multilayer=True` | stacks every level: each triangle becomes a column of wedges, and points carry each level's values |
| Layer thickness | `layer_thickness=10000` | the distance between levels in the multilayer view, in meters on the sphere and degrees on the map; 0 picks one so that the whole stack is easy to see |
| Layers go up (atmosphere) | `atmosphere=True` | stacks levels outwards (or upwards on the map) instead of downwards |
| Layout | `layout="Lat/Lon map"` | `"Sphere"` (default) or a flat map with longitude and latitude in degrees |
| Map center longitude | `center_lon=0` | the longitude in the middle of the map (180 by default) |

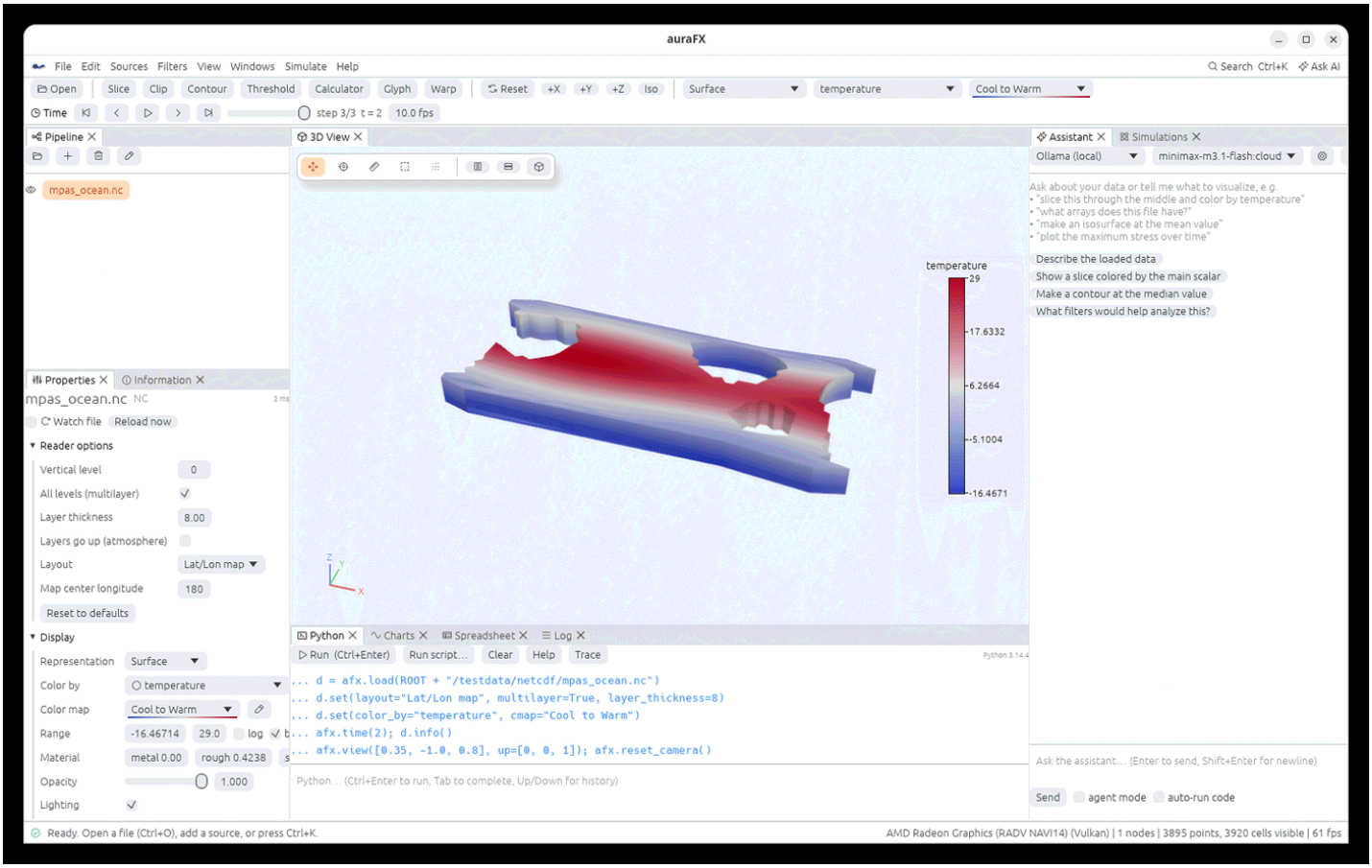
```
ocean = afx.load("output.nc")
ocean.set(vertical_level=10)                          # the 11th level down
ocean.set(layout="Lat/Lon map", multilayer=True)      # all levels, as a flat 3-D
map
```

On the map, cells that cross its left or right edge are drawn on both sides, so the map reaches a little beyond 360°, as in ParaView.

Some things differ from ParaView:

- ParaView adds a point at the planet's center. It collapses onto that point the triangles at a mesh boundary and, when the file has `maxLevelCell`, the triangles that touch land. auraFX leaves those triangles out, so land shows as holes. In the multilayer view, the levels below the sea floor are left out too.
- The default layer thickness of 0 picks a visible spacing; ParaView uses 10000 everywhere.
- ParaView 6.1.1 fills point values in the multilayer view with garbage numbers (around $10^{35}$); auraFX gives each point its level's value.
- Reader options do not reach files opened on another machine through a server yet.






## GIS rasters

A GeoTIFF is a TIFF that also records where it lies on the map. auraFX notices these tags and reads the file as a map raster instead of as a picture. Esri ASCII grids ( `.asc` ), a common text format for elevation models, are read the same way. Open them with **File › Open** or `afx.load("dem.tif")` .

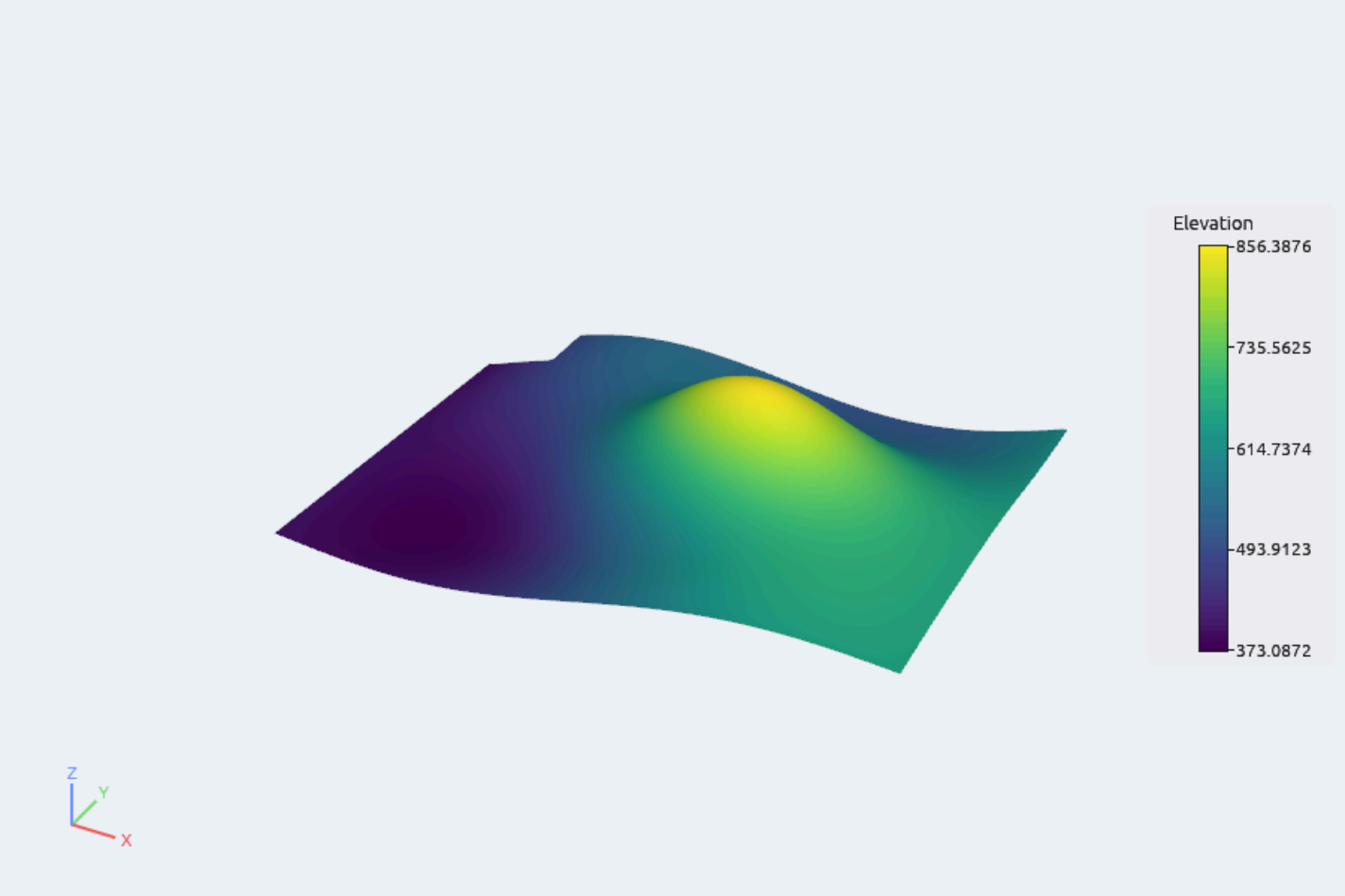
- The raster becomes image data with one cell per pixel, placed at its map coordinates (meters for UTM, degrees for longitude/latitude), with north up.
- Values are cell data. A single band, such as an elevation model, is named `Elevation` , as in ParaView. In a file with several bands, the first is `Elevation` and the others are `Band 2` , `Band 3` … An RGB(A) image becomes one `Elevation` array with 3 or 4 components.
- Pixels equal to the file's no-data value become NaN and are drawn grey (after Warp By Scalar they leave a hole, since a NaN height has no place). ParaView hides those cells instead.
- The field data holds `NO_DATA_VALUE` per band and `MAP_PROJECTION` . For a GeoTIFF that is the EPSG code ( `EPSG:32633` ); ParaView writes a PROJ string instead. For an ASCII grid it is the text of the `.prj` file beside it.
- 8- to 64-bit integer and floating-point bands are read, compressed or not, in strips or tiles, with the bands stored together or one after another. A pixel-is-point GeoTIFF is moved

half a pixel to its corner, as GDAL does.

To see an elevation model as terrain, move the values to the points with **Cell Data to Point Data**, then raise them with **Warp By Scalar**:

```
dem = afx.load("dem.tif")
terrain = dem.cell_data_to_point_data().warp_by_scalar(array="Elevation",
scale=1.0)
terrain.set(color_by="Elevation")
```

Other GDAL raster formats (Erdas `.img`, NetCDF through GDAL, JPEG 2000…) are not read yet.




## GIS vectors

Maps of points, lines and areas, such as countries, roads, rivers, wells or survey sites, open like any other file:

| Format | Files |
|---|---|
| ESRI Shapefile | `name.shp` with `name.dbf` (attributes) and `name.prj` (coordinate system) beside it; open the `.shp` |

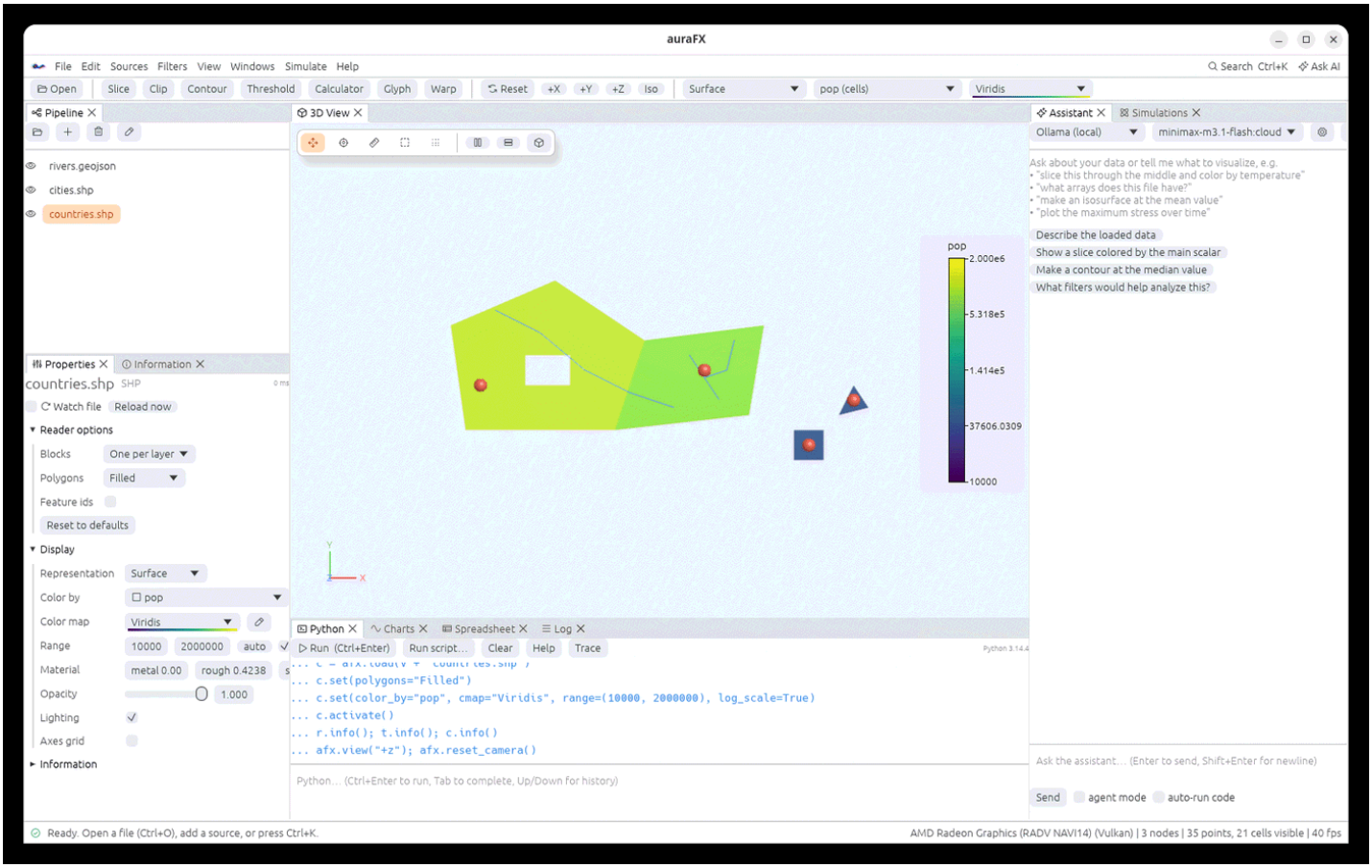
| Format | Files |
|---|---|
| GeoJSON | `.geojson`, or a `.json` file that holds GeoJSON |
| KML (Google Earth) | `.kml`; each folder is a layer |
| GeoPackage | `.gpkg`; each feature table is a layer |

What you get:

- The coordinates stay as stored: longitude and latitude in degrees for most GeoJSON and KML files, or meters for projected shapefiles. Heights (the third coordinate) are kept, so a file with heights in meters on a map in degrees looks very tall. Flatten it with **Transform** (scale z by 0) if that gets in the way.
- Points become vertices, lines become polylines, and polygon rings become closed lines, as in ParaView.
- Each feature's attributes become cell data on its cells: whole numbers as integers, decimals as doubles, and anything else (names, dates) as text. 64-bit integers stay numbers; ParaView turns them into text.
- `MAP_PROJECTION` in the field data holds the coordinate system: the `.prj` text, the EPSG code of a GeoPackage layer, or `EPSG:4326` (longitude/latitude) for GeoJSON and KML.

**Reader options** (Properties › Reader options, or `node.set(...)`):

| Option | Choices |
|---|---|
| **Layer** (files with several layers) | **All** or one layer by name |
| **Blocks** | **One per layer** (default: one block per layer, a row of attributes per feature), or **One per feature (ParaView)**, which is ParaView's layout and slow for thousands of features |
| **Polygons** | **Outlines (ParaView)**, or **Filled**: triangles covering each area, with holes left open, to color areas by an attribute |
| **Feature ids** | adds the feature number as `_vtkPedigreeIds` |




```
countries = afx.load("countries.shp")
countries.set(polygons="Filled")
countries.set(color_by="pop", log_scale=True)
roads = afx.load("world.gpkg")
roads.set(layer="roads")                    # one layer of a GeoPackage
```

Checked against ParaView 6.1.1's GDAL vector reader on shapefiles, GeoJSON and GeoPackage files written by GDAL: with one block per feature, every point, cell and attribute is the same. ParaView reads only the first layer of a file unless told otherwise; auraFX reads all of them. ParaView's GDAL cannot read KML, so the KML reader was checked against the same data saved as GeoJSON.

Not read yet: KMZ (zipped KML), GML, FlatGeobuf, MapInfo, file geodatabases, and curved geometry (arcs). Shapefile multipatches (3-D surfaces) are skipped, as ParaView does.




## AMR data (adaptive mesh refinement)

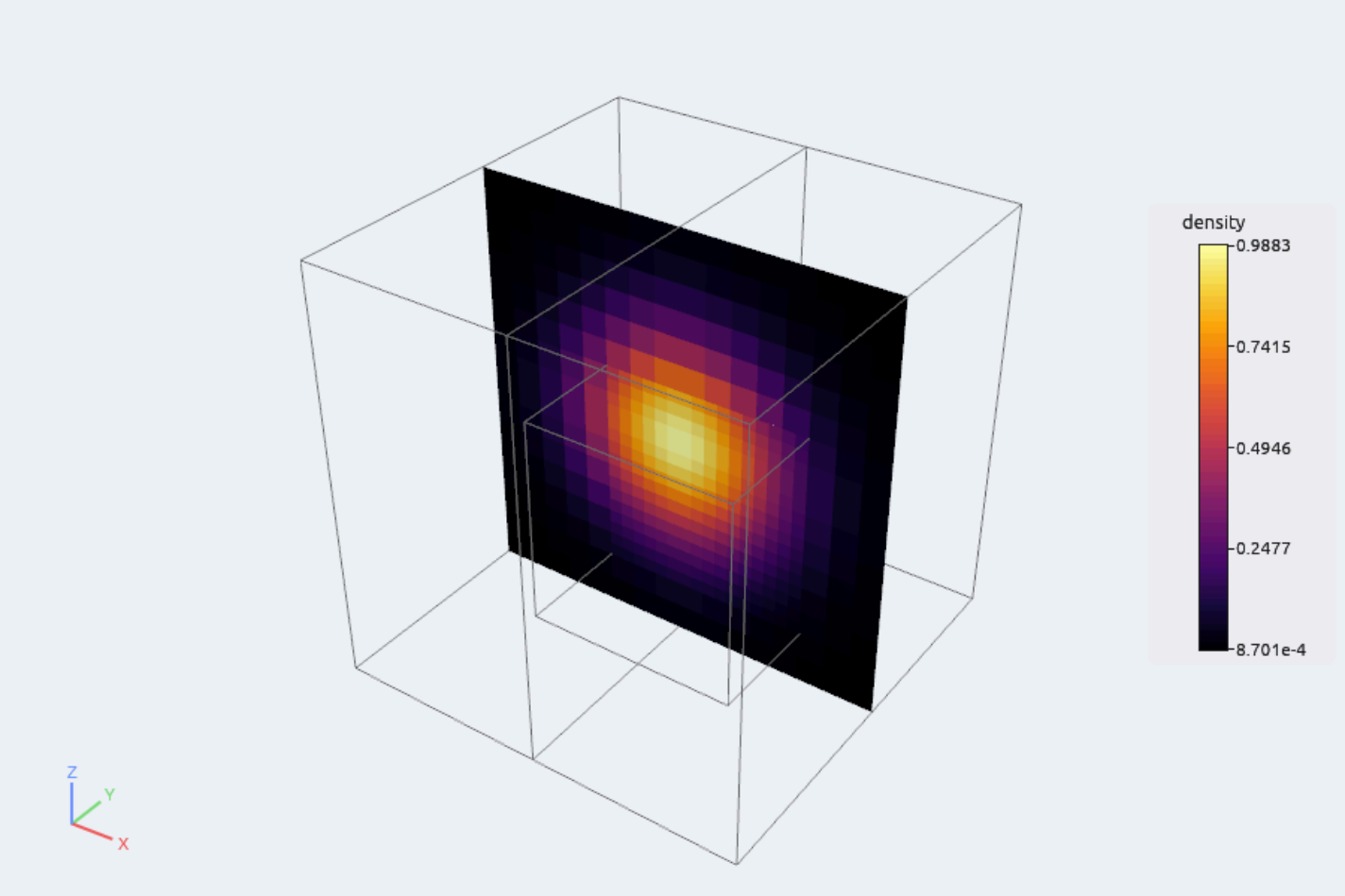
AMR simulations (AMReX, BoxLib, Castro, MAESTRO, Nyx, WarpX…) store their domain as levels of boxes: a coarse level everywhere and finer boxes where more detail was needed. auraFX reads:

- **AMReX / BoxLib plotfiles**: a folder like `plt00100` with a `Header` file. Open the `Header` file in **File › Open** (choose "All files"), drop the folder on the window, or `afx.load("plt00100")`.
- **VTK AMR files** (`.vthb`, `.vth`), as ParaView writes them.

What you get:

- One block per level (**Level 0**, **Level 1**, …), each holding that level's boxes as image data with the variables as cell data. The **Information** panel lists them.
- Coarse cells that a finer box covers are marked as hidden in `vtkGhostType`, as in ParaView. Surfaces, slices, clips, contours, thresholds, Integrate Variables, Resample To Image and probing skip them, so every place is shown and counted once, at its finest level.
- AMReX vector variables written as `amrexvec_<name>_x`, `_y`, `_z` become one vector `<name>`. In 2-D files the missing z is 0. ParaView puts the file's first variable there instead.

- The plotfile's time is in the field data ( `TimeValue` ). Single- and double-precision plotfiles are read.
- **Outline** draws the outline of every box, which shows where the refinement is.




```
amr = afx.load("plt00100")                # the plotfile folder
amr.set(representation="Outline")         # where the refinement is
s = amr.slice(normal=(0, 0, 1))           # finest data on the cut
s.set(color_by="density")
lev = amr.overlapping_amr_level_ids()     # LevelIdScalars: each cell's level
```

**Sources › AMR Gaussian Pulse** makes ParaView's small AMR test dataset: a coarse box and two finer boxes holding a Gaussian pulse, in 2-D or 3-D. **Filters › Overlapping AMR Level Ids** adds the cell array `LevelIdScalars` with each cell's level.

Checked against ParaView 6.1.1: the AMR Gaussian Pulse source (cells, values, covered cells), the `.vthb` reader on files ParaView wrote, the AMReX reader on test plotfiles ( `scripts/gen_amrex.py` ; every box's origin, size, sums and covered cells), and on the AMR data Integrate Variables (the volume counts each place once), slices, Threshold, Extract Surface, Cell Data to Point Data and Resample To Image.

Not read yet: plotfiles without FAB headers (the "NoFabHeader" variants), AMReX particles, Enzo, FLASH and Chombo files. Not available yet: ParaView's AMR-specific filters (AMR Contour,

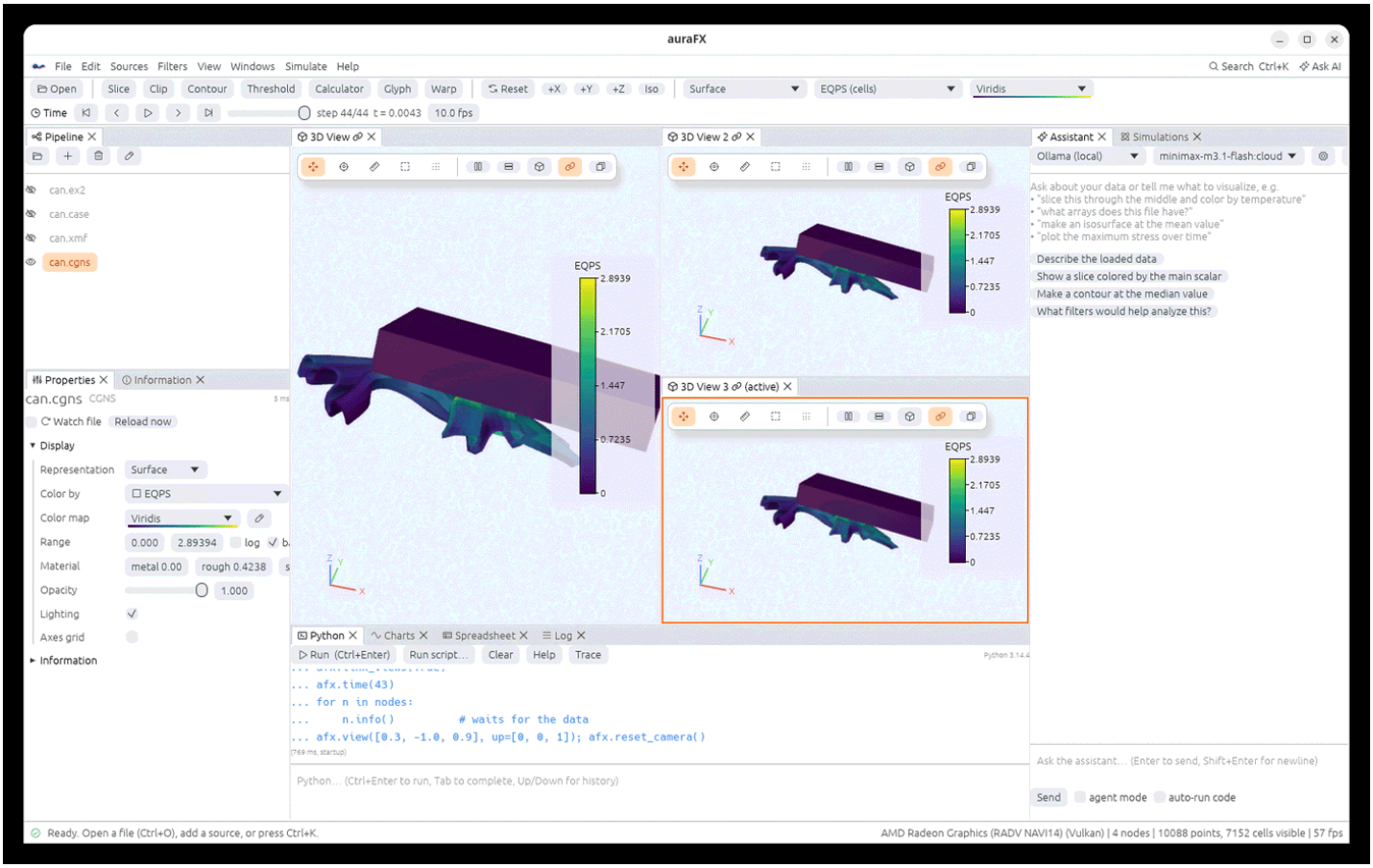
Dual Clip, Fragments, Resample AMR).

## Writers

| Format | Extensions | Notes |
|---|---|---|
| VTK XML | `.vtu` `.vtp` `.vti` `.vtr` `.vts` `.vtm` | appended binary; multiblocks as a folder of pieces |
| VTK legacy | `.vtk` | |
| VTKHDF | `.vtkhdf` | all dataset kinds; multiblocks as a collection; time series with `all_times=True`; written with auraFX's own HDF5 encoder (read by h5py and VTK) |
| Exodus II | `.exo` | blocks, nodal and element variables, every time step with `all_times=True` |
| EnSight Gold | `.case` | C binary; one part per block (image, rectilinear and structured grids as blocks, the rest as elements of every standard type, polygons and polyhedra); every time step with `all_times=True`, the geometry written once when it does not change |
| CGNS | `.cgns` | HDF5 storage (CGNS 4): a zone per block, image/rectilinear/structured grids as structured zones, the rest as one element section (MIXED when the cell types differ) plus NGON/NFACE for polygons and polyhedra; vectors as `NameX/NameY/NameZ`; with `all_times=True` one solution per step (and coordinates per step when the points move), but the cells must stay the same |
| Xdmf 3 | `.xmf` + `.h5` | the description in the `.xmf`, the arrays in an HDF5 file beside it; block trees as spatial collections, time steps as a temporal collection that shares unchanged points and cells |
| CSV, STL, PLY | | |
| Scenes | `.glb` `.obj` `.x3d` `.wrl` | see Exporting |

| Format | Extensions | Notes |
| --- | --- | --- |
| Pictures | `.png` `.svg`, GIF/MP4 animations | see Exporting |

Files written by the VTK XML, VTKHDF, Exodus, EnSight, Xdmf and CGNS writers open in ParaView with the same points, cells and values. EnSight groups the cells of each part by type, and CGNS puts polyhedra after the other cells, so cells can come back in a different order (each keeps its values). Cell types a format does not have (Lagrange and Bezier cells, for example) are written by their corner points. A CGNS zone holds cells of one dimension: in a volume mesh, boundary faces and lines are left out.




## Curved cells

VTK's Lagrange and Bezier cells of any order, quadratic cells and serendipity cells are read with their node numbering, drawn curved and cut by the filters along their curvature; see Volumes, point clouds and curved cells.

## Your own formats

A plugin reader is a Python function with `@afx.reader(extensions=[...])`; it takes precedence over the built-in reader for its extensions and appears in the Open dialog.

# Exporting pictures, scenes and animations

## Screenshots

**File › Save Screenshot…** (Ctrl+Shift+S) opens a small dialog:

- **Size**: the view's size, a multiple of it (2×, 4×), 1920×1080, 3840×2160, or any size; **keep the view's shape** ties width and height together. Point sizes and line widths are in pixels, so they look thinner in a larger picture.
- **Background › transparent** saves a PNG with an alpha channel, with clean edges.
- **Annotations** includes the legend, axes-grid labels, time label, text annotations and the orientation axes, laid out as on screen and scaled with the picture.
- **PNG** or **SVG**: an SVG holds the surfaces as shaded polygons sorted back to front, the lines and points, and the annotations as real text, for figures that scale without pixels (where surfaces cut through each other the boundary is jagged; convert to PDF with Inkscape if needed).

With ray tracing on, the picture is traced at the requested size first.

```
afx.screenshot("fig.png")                                  # the view as shown
afx.screenshot("fig.png", width=3840, transparent=True) # keeps the view's shape
afx.screenshot("fig.png", width=1600, height=900, annotations=False)
afx.screenshot("fig.svg")
```

`afx.screenshot` returns only when the file is written, so a script can change the camera or the data between pictures.

## Scenes for other programs

**File › Export Scene…** ( `afx.export_scene("scene.glb")` ) writes the visible objects, with their colors, as:

| Format | Notes |
|---|---|
| **glTF binary** ( `.glb` ) | for Blender, three.js, PowerPoint, Windows 3D Viewer; vertex colors, normals, translucency, materials (metallic, roughness) |
| **OBJ** ( `.obj` ) | with vertex colors after the coordinates (Blender and MeshLab read them) |

| Format | Notes |
|---|---|
| **X3D** ( `.x3d` ), **VRML** ( `.wrl` ) | absolute coordinates, vertex colors; ParaView reads the VRML |

Volumes are not exported (they are not geometry).

## Animations

**File › Animation…** records the time steps, a turntable or the keyframes as **GIF**, **MP4** ( `ffmpeg` on the PATH) or a folder of PNG frames; see Time, charts and animation.

## Saving data

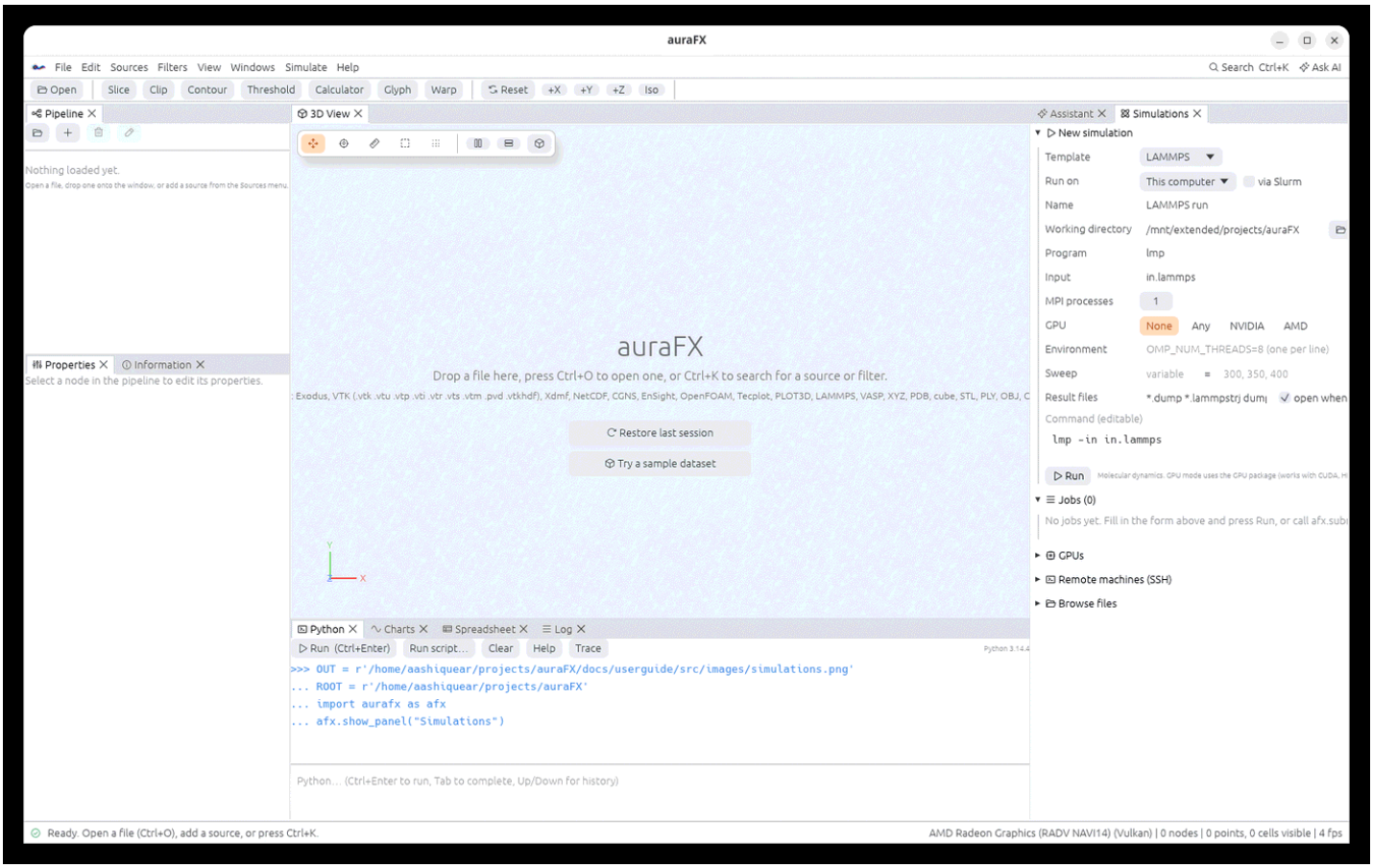
Right-click an object › **Save data…**, or `node.save("out.vtu")` , writes its output; the format follows the extension:

| Extension | Format |
|---|---|
| `.vtu .vtp .vti .vtr .vts .vtm` | VTK XML (appended binary); `.vtm` writes a folder of pieces |
| `.vtk` | legacy VTK |
| `.vtkhdf` | VTKHDF (HDF5); multiblocks as a collection with the block tree; `all_times=True` writes every time step into one file |
| `.exo` | Exodus II with blocks and variables; `all_times=True` writes every time step |
| `.case` | EnSight Gold (C binary): the case file plus a geometry file and one file per variable next to it; `all_times=True` writes every time step |
| `.xmf` | Xdmf 3: the description plus an `.h5` file with the arrays; `all_times=True` writes every time step |
| `.cgns` | CGNS (HDF5); `all_times=True` writes every time step when the cells stay the same (moving points are fine) |
| `.csv` | the point data or the table |
| `.stl .ply` | surfaces |

```
d.save("slice.vtp")
d.save("run_all.exo", all_times=True)
d.save("run_all.vtkhdf", all_times=True)
d.save("run.case", all_times=True)      # run.case, run.geo, run.Temp0000, ...
d.save("run.xmf", all_times=True)       # run.xmf and run.h5
d.save("run.cgns", all_times=True)
```

When the data has several time steps and the chosen format holds them all (Exodus, VTKHDF, EnSight, Xdmf, CGNS), the Save dialog writes every step.

# Running simulations




The **Simulations** panel (right side; **Simulate › Run a simulation…**, or **Windows › Simulations**) launches solvers and opens their results when they finish.

## A new simulation

| Field | Meaning |
|---|---|
| **Template** | LAMMPS, MOOSE app, OpenFOAM, Python pipeline or Custom command; fills in the program, input and result-file patterns |
| **Run on** | this computer, or a remote machine from the SSH list; **via Slurm** submits with `sbatch` there (partition, time and extra `#SBATCH` lines appear) |
| **Name**, **Working directory** | where the run lives; a remote directory when running over SSH |
| **Program**, **Input**, **MPI processes** | the command line is assembled from these and shown below, editable |

| Field | Meaning |
|---|---|
| **GPU** | None, Any, NVIDIA or AMD; the panel finds the GPUs on the host and sets the solver's flags (LAMMPS GPU package, `CUDA_VISIBLE_DEVICES` / `HIP_VISIBLE_DEVICES`) |
| **Environment** | extra `VAR=value` lines (one per line) |
| **Sweep** | a variable name and a list of values: one job per value, `{var}` in the command replaced, each in its own folder |
| **Result files** | patterns of the files to open when the run ends (**open when done**) |

**Run** starts the job. Jobs appear in the **Jobs** list with their state, a live log tail, and buttons to cancel, open the results or the folder, and resubmit. Files a running solver writes can be opened while it runs; auraFX reloads them as they change (live reload).

## Remote machines

**Remote machines (SSH)** holds the hosts: a name, the SSH target (`user@machine`, or an alias from `~/.ssh/config`), port, key file, jump host, and **setup lines** run before every command (module loads, `source` of an environment, `AURAFX_SERVER=...`). SSH keys or an agent must be set up so that `ssh host` works without a password prompt. The same hosts serve **File › Open on Server...** (see Data on another machine).

**GPUs** lists the GPUs of each host; **Browse files** opens a remote file browser to pick inputs or results.

## From Python

```
jid = afx.submit("lmp -sf gpu -pk gpu 1 -in in.melt", host="rocm-server",
                 workdir="~/runs/melt", gpu="amd:0", outputs="dump.melt",
auto_load=True)
afx.sweep("lmp -var T {T} -in in.melt", T=[300, 600, 900], host="cluster",
workdir="~/sweep")
afx.jobs(); afx.job(jid); afx.cancel(jid)
afx.gpus("rocm-server"); afx.hosts()
afx.load_remote("rocm-server", "~/runs/melt/dump.melt")   # copy a file here and
open it
```

`gpu` is `None`, `"any"`, `"nvidia"`, `"amd"` or e.g. `"nvidia:0,1"`. `afx.job(id)` returns the state, exit code, log tail and output files.

# Data on another machine

When the data lives on a cluster or a GPU box, auraFX can read and filter it **there** and bring back only what you look at, like ParaView's `pvserver`, with nothing to set up beyond SSH.

## How it works

`aurafx-server` is a small program without a window. The app starts it on the host over SSH (`ssh host aurafx-server --stdio`) and talks to it through that connection, so keys, jump hosts and `~/.ssh/config` work as usual and no port is opened. Files you open on the server are read there; filters you add below them (Slice, Clip, Contour, Threshold, Stream Tracer, temporal filters and so on) run there too. What comes back:

| Situation | What is transferred |
|---|---|
| Visible, output smaller than 64 MB | the whole output |
| Visible, larger | its outer surface (all arrays), which is what the 3D view draws |
| Hidden (for example a reader with a slice below it) | only its summary: arrays, ranges, bounds, time steps |
| **Bring all data** ticked in Properties, or a filter below it that runs here | the whole output |

The Information panel always describes the full data on the server. Python filters (Programmable Filter, Python Calculator) and plugin filters run on this computer, on the data brought over. The server keeps results, so changing one filter only re-runs that filter there.

## Setting up a host

1. Make sure `ssh yourhost` works without a password prompt.
2. Put `aurafx-server` on the host: **File › Open on Server… › Install server** copies the one shipped with the app to `~/.local/bin` (Linux hosts of the same architecture; the packages run on glibc 2.35 or newer), or build it there with `cargo build --release -p aurafx-server`, or point `AURAFX_SERVER` at it in the host's setup lines.
3. **Check server** shows its version and GPU.

Hosts come from the Simulations panel and from `~/.ssh/config`. “This computer” runs the server locally, which is handy for trying it out.

## Opening files

**File › Open on Server…**: pick the host, browse to the file, double-click it. A cloud icon marks objects that run on a server; hover it, or look at the top of Properties, to see what was transferred and how long the server took.

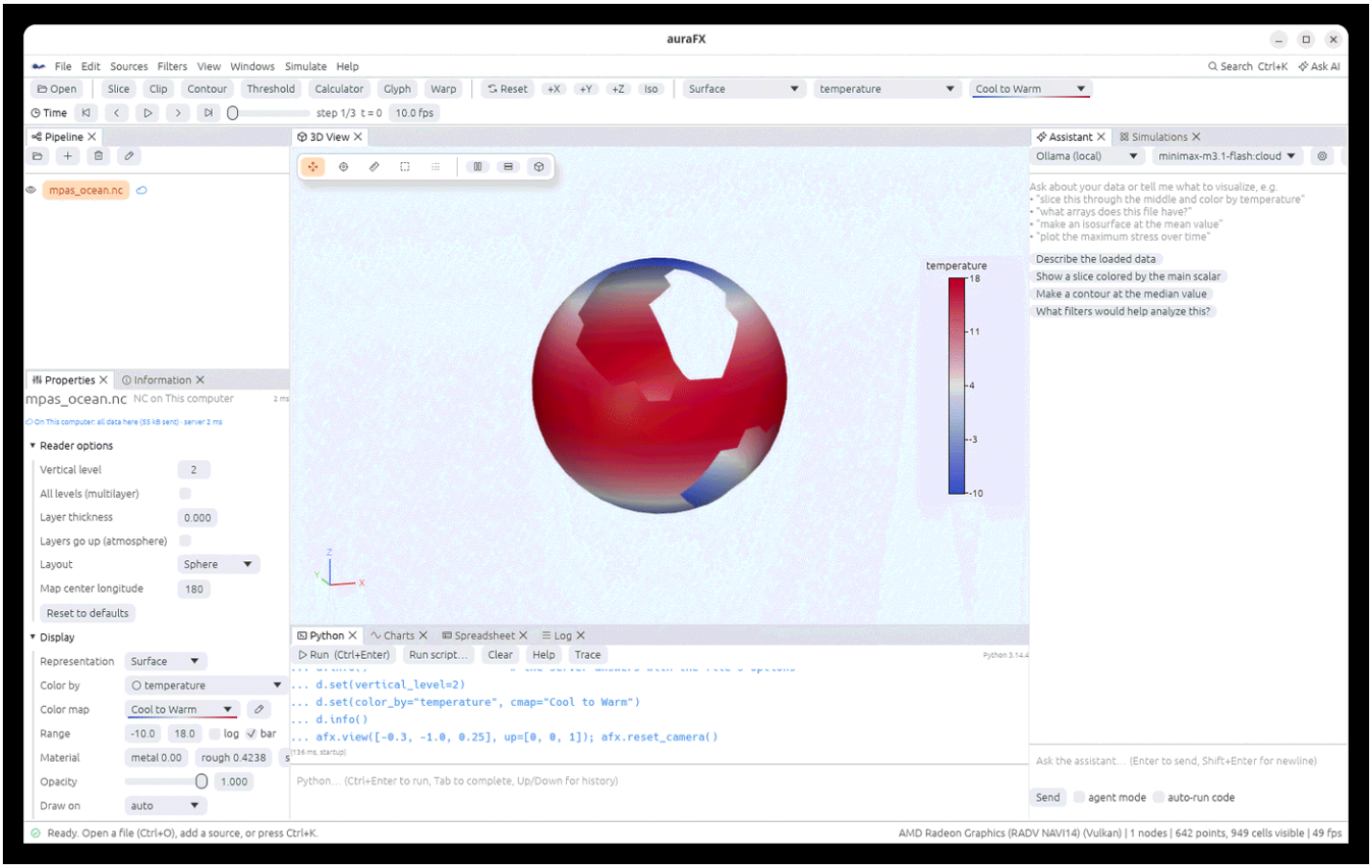
```
d = afx.load("/scratch/run42/out.e", host="cluster")   # a Simulations host, ssh
alias or user@machine
d = afx.load("ssh://cluster/scratch/run42/out.e")       # the same
s = d.slice(normal=(0, 0, 1))                            # runs on the server
s.set(all_data=True)                                     # bring the whole output
over
afx.servers()                                            # connections and the
machines' details
```

### Reader options on the server

Files on a server have the same **Reader options** as local ones (an MPAS file’s vertical level and layout, a NetCDF file’s grid, a CAM-SE file’s connectivity file…). The server looks at the file and reports its options with its first answer, so they appear under Properties a moment after the file opens. Changing one has the server read the file again with it; on a parallel server, every process reads its piece with it. Paths in options, such as a CAM-SE connectivity file, are paths on the server.

```
d = afx.load("/scratch/ocean/output.nc", host="cluster")
d.set(vertical_level=10)          # read on the server
```

From Python, an option the file does not have is refused once the server has said which ones it has; before that, the server ignores names it does not know. Saved states keep the options.




## Drawing on the server

Some outputs are too big even as a surface. Then the server draws the object with its own GPU and sends pictures (a color and a depth image per view, compressed), which the 3D view blends by depth with the local objects, so everything hides and is hidden correctly. While you move the camera the last picture stays up until the next one arrives.

**Display › Draw on** in Properties picks per object: **auto** (drawn here unless the surface is over 64 MB), **this computer**, or **the server** ( `node.set(draw_on="server")` ). Picking, rulers and selections need the geometry here, so switch to “this computer” for those. Translucent objects, volumes and Surface LIC are always drawn here. The view’s lights and shadows travel with each request; screen effects (ambient occlusion, Eye-Dome Lighting, FXAA, tone mapping) are applied here to the whole picture.

## Parallel servers

Data too big for one process can be spread over several, like `mpirun pvserver` : **Open on Server › Processes**, or `afx.load(path, host="cluster", processes=8)` , starts eight worker

processes on the host (`aurafx-server --workers node1:16,node2:16` spreads them over machines sharing a file system). Partitioned VTK files are read piece by piece; other files are read by every process and split. Cell-by-cell filters run on every piece with automatic values taken from the whole data; the pieces come back appended, as a merged surface, or as pictures composited by depth. Results match a single process exactly.

Not yet: MPI launchers (`srun`) for the workers, ghost cells, sharing one server between people, temporal filters on parallel servers.

# Settings and keyboard shortcuts

## Settings

**Edit › Settings…** (Ctrl+,):

| Setting | Meaning |
|---|---|
| **Theme** | dark, light or system; the 3D background per theme |
| **UI scale** | enlarges the whole interface |
| **Orientation axes**, **Time label** | shown in the 3D views |
| **Filter threads** | cores used by the parallel filters (0 = all); takes effect at the next start; `AURAFX_THREADS` overrides it |
| **Default color map** | for new objects |
| **AI** | provider, model, server URL, API key, auto-run and agent mode |
| **Plugins** | the Plugins window (also **Edit › Plugins…**) |
| **Lighting** | in **View › Lighting**; the choices are saved with the settings |

Settings live in `~/.config/aurafx/settings.toml` (Linux) or the app's config folder on other systems; the window layout in `~/.local/share/aurafx/app.ron`.

## Keyboard shortcuts

| Keys | Action |
|---|---|
| Ctrl+O | open a file |
| Ctrl+S | save state |
| Ctrl+Shift+S | save a screenshot |
| Ctrl+K, Ctrl+Space | command palette |
| Ctrl+Z, Ctrl+Y | undo, redo |
| Ctrl+Enter | run the code in the Python console |
| Ctrl+, | settings |

| Keys | Action |
|---|---|
| R | reset the camera |
| Esc, P, M, S, D | view modes: navigate, pick, ruler, select cells, select points |
| Delete | delete the selected object |
| Space | play / pause time |
| Double-click in the view | fit the data |

**Help › Keyboard shortcuts** shows this list in the app.

## Environment variables

| Variable | Use |
|---|---|
| `AURAFX_THREADS` | thread limit for filters |
| `AURAFX_PLUGIN_PATH` | extra plugin folders |
| `AURAFX_SERVER` | path of `aurafx-server` on a host (in its setup lines) |
| `AURAFX_DATA_DIR` | where settings and state are kept |
| `OLLAMA_HOST`, `ANTHROPIC_API_KEY` | AI providers |
| `PYO3_PYTHON` (at build time) | which Python to embed |